Resolution refinement followed by the rodeo algorithm for eigenstate preparation: implementation and assessment on the pairing Hamiltonian

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Projective spectral filters such as the rodeo algorithm prepare eigenstates with exponential accuracy in the number of filter cycles, but their cost is set by the squared overlap p of the input state with the target. We implement and assess, on the exactly solvable constant-pairing Hamiltonian, a pipeline in which *resolution refinement* — the adiabatic transport of an eigenstate from a coarse to a fine single-particle basis along a shift-controlled, gapped path — supplies the input of the rodeo filter, and we compare it against the simplest alternatives on the same footing: the same filter applied to the unrefined prolonged state and to a full-space unitary coupled-cluster (UCCSD) state. All comparisons use the ground-state infidelity as the common observable, the same second-order product-formula evolution, and a two-qubit gate budget per accepted preparation that includes the embedded coarse-Hamiltonian term of the refinement path. For four particles on four levels (eight system qubits) the refined input ($p = 0.998$) is filtered to an infidelity of 5×10^{-4} after two cycles and 1.5×10^{-4} after six, close to the floor set by the Trotterized evolution; the same filter brings the prolonged determinant ($p = 0.868$) to a comparable accuracy at a comparable gate count, while the UCCSD state, whose actual infidelity is 1.8×10^{-4} although its energy error is 1.4×10^{-3}, reaches 1.3×10^{-4} after two cycles at one third of the cost. Refinement therefore does not beat a cheap, accurate variational input at this system size; its merits are the absence of a classical optimization loop and of training measurements. The extracted refinement time grows with the inverse of the minimum path gap when the gap is varied through the energy shift, and with the loss of initial overlap when the pairing strength is varied. A pair-breaking particle-hole interaction, which destroys the seniority symmetry of the model, leaves the structure and convergence of the pipeline unchanged, raises the depth of a Trotter step ninefold, and leaves the hybrid tied with the filtered UCCSD state at weak and behind it at strong pair breaking. Level occupation numbers and correlation energies of the prepared states are reported, and all stages are compiled to rotation–CNOT circuits with gate counts verified against exact evolution.

I. INTRODUCTION

Determining the low-lying eigenvalues and eigenvectors of a many-body Hamiltonian is one of the grand challenges of quantum many-body theory, where the Hilbert-space dimension grows exponentially with the number of constituents. Quantum computers offer new routes to eigenvalue estimation and eigenstate preparation, and their prospects for nuclear structure have been mapped in recent community studies [12, 55, 56], with schematic nuclear models — pairing, Lipkin, and shell-model Hamiltonians — serving as the standard proving ground [41–43, 45–48]. The state-of-the-art preparation methods, however, each face an obstacle at scale. The variational quantum eigensolver (VQE) [1] is limited by the cost of its classical optimization loop, by the accuracy floor of its ansatz and, for sufficiently expressive circuits, by barren plateaus — gradients that vanish exponentially with system size [2, 54]. Adiabatic state preparation requires evolution times that diverge if the gap between the tracked state and the rest of the spectrum closes along the path [3]. Projective spectral filters — quantum phase estimation [4], polynomial filters based on quantum signal processing [52, 53], projection by measurement [44], and the rodeo algorithm (RA) [5, 6, 49–51] — converge exponentially in the number of cycles, but only once the initial state has a significant squared overlap p with the target eigenstate; their cost grows as $1/p$, and for a fixed, unoptimized reference state p falls exponentially with system size.

These limitations are complementary, and the idea of *preconditioning* the input of a projective filter by an adiabatic stage is as old as the rodeo algorithm itself: the orig-

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inal proposal [5] already combines a short adiabatic ramp with the filter, the hierarchical fusion method of Ref. [8] builds large systems from exactly solvable subsystems by alternating adiabatic ramps and rodeo cycles, and the resolution-refinement proposal [7] explicitly recommends switching from adiabatic evolution to a filter such as the rodeo algorithm once the overlap is large. *Resolution refinement* bootstraps an eigenstate across model-space scales: an eigenstate of a coarse (low-resolution) Hamiltonian is lifted by a prolongation operator into a finer space and adiabatically evolved to the fine-resolution eigenstate. Because refinement changes the low-energy spectrum only mildly, the interpolation path can be kept gapped, and under the additional assumption that the first-order perturbative correction to the tracked state stays bounded [7] the required adiabatic time scales as the inverse gap times a slowly growing system-size factor rather than as the inverse gap squared. The same bootstrapping idea applied along the other axis is *nucleation*, proposed recently by some of us [9]: the exact ground state of a small lattice (the seed) is compiled into a circuit, and the lattice is then grown in stages, each stage a Trotterized adiabatic interpolation that couples newly added qubits to the existing bulk, until the target size is reached. Refinement keeps the domain fixed and raises the resolution; nucleation keeps the resolution fixed and grows the domain. Both replace one hard global preparation by a sequence of local, gapped adiabatic steps that start from an exactly solved problem, both are compiled as Trotterized adiabatic circuits, and Ref. [9] supplies for such circuits sufficient conditions on the evolution time and on the number of Trotter steps, together with a variational variant that reuses the Trotterized circuit as a warm-started ansatz. The present work implements this chain — (i) prepare a low-resolution eigenstate; (ii) prolong and adiabatically refine for a time T; (iii) apply the rodeo algorithm — for a hierarchy of fermionic single-particle bases, and asks the question that decides whether the refinement stage earns its gates: does refinement followed by filtering beat filtering the best input that is available cheaply?

The pairing problem has become a standard target for quantum algorithms, and the programme of Lacroix and collaborators is the closest in spirit to ours. Its common thread is the deliberate use of symmetry breaking and restoration: a symmetry-breaking (BCS-like) trial state is cheap and correlated, and the symmetry is restored afterwards by a projective step — quantum phase estimation used as a projector [23], Grover-type oracles [28], spin filters [25] or classical shadows [31] — with the restored state then handed to a spectral post-processing stage such as phase estimation or a Krylov expansion [24, 26]; Refs. [27, 48] review this viewpoint and Ref. [29] the broader many-body landscape. More recently the same group has used ADAPT-VQE on pairing Hamiltonians with symmetry-conserving and symmetry-breaking operator pools [47], extracted excited states from the subspace spanned along the ADAPT convergence path [32], obtained Lipkin spectra with a quantum equation-of-motion approach [42] and with a two-qubit generator-coordinate scheme [30], and built one-body Green's functions of small superfluid systems from a variational even-N ground state and a quantum subspace expansion for its $N\pm1$ neighbors [33]. Our pipeline addresses the step that all of these share — obtaining a high-fidelity eigenstate of the fine-resolution Hamiltonian from a cheap approximate one — with a different mechanism: not symmetry projection or subspace diagonalization, but an adiabatic path across resolutions followed by an energy filter; the relation is taken up in Sec. VII.

In this work we realize the full pipeline end to end on a transparent and exactly solvable testbed, the constant-pairing Hamiltonian familiar from nuclear and condensed-matter many-body theory [21] and integrable by the Richardson ansatz [38–40]. Its exact solvability is precisely what makes it the right laboratory: every stage of the pipeline can be validated quantitatively — the gate-level UCCSD coarse solver against full configuration interaction (FCI) and coupled-cluster doubles (CCD); the prolongation and adiabatic refinement against the exact high-resolution eigenstate; and the rodeo filter against the analytic success-probability formulas of Ref. [6]. All circuits are implemented explicitly as rotation and CNOT gates (and an ancilla-based rodeo cycle), so the demonstration doubles as a reference implementation for hardware.

Relative to the resolution-refinement proposal of Ref. [7], the rodeo literature [5, 6] and the fusion method [8], which established adiabatic preconditioning of the filter on lattice and spin systems, the present work contributes five elements: (i) an implementation of refinement for a hierarchy of *single-particle bases* in a fermionic pairing problem, where the prolongation is gate-free and the embedded coarse Hamiltonian is a conditioned operator whose circuit we give and count; (ii) a quantitative benchmark of the refinement-plus-filter chain against the same filter applied to the unrefined input and to a full-space UCCSD state, with the ground-state infidelity as the common observable and matched evolution accuracy; (iii) a gate-level compilation of every stage — Trotterized refinement and Trotterized controlled evolution included — with a two-qubit-gate budget per accepted preparation that separates one-time training, repeated preparation and the filter; (iv) a test of the pipeline on a seniority-breaking Hamiltonian — the pairing model supplemented by a particle-hole interaction [22], which carries the one-particle–one-hole channel of realistic interactions — including the corresponding product-formula study and resource table; and (v) a reproducible reference implementation in which every figure and number derives from a single executable notebook. The outcome of (ii) is stated plainly: at the system sizes accessible to exact validation here, refinement followed by filtering does not outperform filtering a UCCSD input, and is roughly tied with filtering the unrefined prolonged state at matched fidelity; the case for the refinement stage rests

on its freedom from classical optimization and training measurements, and on regimes — larger spaces, poorer variational inputs — that lie beyond what we can validate exactly.

II. THE PAIRING MODEL AND ITS QUBIT ENCODING

We study the constant-pairing Hamiltonian

$$H = \delta \sum_{p=0}^{k-1} p \sum_{\sigma=\uparrow\downarrow} n_{p\sigma} - \frac{g}{2} \sum_{p,q=0}^{k-1} a^\dagger_{p\uparrow} a^\dagger_{p\downarrow} a_{q\downarrow} a_{q\uparrow}, \tag{1}$$

with k doubly degenerate single-particle levels of spacing δ (set to $\delta = 1$) and a pairing strength g. The interaction scatters time-reversed pairs between levels; it conserves the seniority (the number of unpaired particles), so the ground state of N paired particles lives entirely in the seniority-zero subspace, and the model is exactly solvable by the Richardson ansatz [38–40] — the property that anchors every benchmark below. We map level p and spin σ to qubit $2p + \sigma$ (so a k-level problem uses $2k$ qubits) and encode the fermionic operators through the Jordan–Wigner transformation. The Hartree–Fock (HF) reference fills the lowest $N/2$ levels, with $E_{\mathrm{HF}}(N{=}4, g) = 2 - g$.

Throughout we use $N = 4$ particles as the running example. The fixed-N sector dimension is $\binom{2k}{N}$, growing from 1 at $k = 2$ (a single filled determinant) to 70 at $k = 4$ and 1820 at $k = 8$; the seniority-zero ground-state problem is correspondingly compact, which makes the model an ideal multi-resolution laboratory. "Basis refinement" here means *increasing k at fixed N*: the number of single-particle levels (and qubits) grows while the particle number is held fixed.

III. THE HYBRID PIPELINE

A. Coarse-state preparation: gate-level UCCSD–VQE

The coarse eigenstate is prepared with a unitary coupled-cluster ansatz restricted to singles and doubles (UCCSD) [13, 17], Trotterized into a product of excitation circuits acting on the HF reference,

$$|\Psi(\boldsymbol{\theta})\rangle = \prod_{\mu} e^{\theta_\mu (\hat{\tau}_\mu - \hat{\tau}^\dagger_\mu)} \, |\mathrm{HF}\rangle \, . \tag{2}$$

Through the Jordan–Wigner map each anti-Hermitian generator becomes a sum of commuting Pauli strings, so every factor compiles into a basis change, a CNOT ladder, a single $R_z(\theta)$ rotation, and the uncompute; a single excitation yields two Pauli-rotation sub-circuits and a double excitation yields eight (the circuits are shown in Fig. 10). We optimize with the exact analytic gradient of the product ansatz (the naive two-term parameter shift is incorrect beyond the first factor, because later factors acquire frequency-one cross terms). Because the pairing force does not couple to seniority-changing single excitations, all singles amplitudes vanish to machine precision and the solver is effectively a *doubles* theory, mirroring the classical CCD structure.

B. The prolongation operator

The prolongation P embeds a low-resolution state into the high-resolution space. For basis refinement it maps every many-body basis state of the k_{low} space to the *same* occupation pattern in the k_{high} space, leaving the new single-particle levels empty. In canonical Jordan–Wigner ordering this is gate-free; for lattice refinement the analogous operator distributes each occupied coarse site over the neighboring fine sites with two single-qubit rotations and two CNOTs per site per spatial direction, i.e. $O(N_{\mathrm{sites}})$ gates per dimension [7]. The prolonged state $|\Phi_0\rangle = P\,|\psi_{\mathrm{low}}\rangle$ satisfies $\langle\Phi_0|H_{\mathrm{high}}|\Phi_0\rangle = E_{\mathrm{low}}$ exactly, and already provides a high-quality first guess for the high-resolution ground state.

C. Resolution refinement (adiabatic phase)

The prolonged state is refined by adiabatically following the interpolated, energy-shifted Hamiltonian

$$H(s) = \cos^2\!\left(\tfrac{\pi s}{2}\right) P(H_{\mathrm{low}} - \mu)P^\dagger + \sin^2\!\left(\tfrac{\pi s}{2}\right)(H_{\mathrm{high}} - \mu), \qquad s = \frac{t}{T}, \tag{3}$$

evolved by a Suzuki–Trotter product over total time T. The shift μ (here $\mu = E_{\mathrm{low}} + 0.6$) places the tracked low-energy states below the large null space of $P^\dagger$, so the level

followed from $s = 0$ to $s = 1$ is the high-resolution ground state. The first term is *not* the coarse Hamiltonian acting on the coarse register alone: since $PP^\dagger$ projects on the states in which the appended single-particle levels are empty,

$$A \equiv P(H_{\rm low}-\mu)P^\dagger = (H_{\rm low}-\mu)\otimes|0\cdots0\rangle\langle0\cdots0|_{\rm app}\,, \quad (4)$$

a coarse-space operator conditioned on the $2(k_{\rm high}-k_{\rm low})$ new qubits being in $|0\rangle$; its circuit and cost are given in Sec. V B. At $s = 0$ the gap above the tracked state is $\min(E_1^{\rm low} - E_0^{\rm low},\, \mu - E_0^{\rm low})$: increasing the shift lowers the embedded coarse states further below the null space, and the gain in initial gap saturates at the coarse excitation gap (for a one-dimensional coarse space, as in the $k = 2$ start used below, it is simply $\mu - E_{\rm low}$). Along the path the instantaneous spectrum is found to stay smooth and gapped in all cases studied here. A gapped path alone does not fix the adiabatic time scale; Ref. [7] shows that when the squared norm of the first-order perturbative correction to the tracked state stays bounded along the path — a condition on the transition matrix elements of dH/ds relative to the excitation gaps — the adiabatic time scales as $T \sim 1/\Delta E$ rather than as the $O(1/\Delta E_{\rm min}^2)$ of the generic adiabatic theorem [3]. We test this empirically in Sec. IV D and treat the overlap $p = |\langle\Psi_{\rm high}|\Phi(T)\rangle|^2$ as the figure of merit of the stage.

The interpolation (3) is the $\sin^2(\pi s/2)$ schedule, whose vanishing derivatives at both ends remove the boundary terms of the adiabatic bound of Jansen, Ruskai and Seiler [10]; it is also the schedule of the nucleation method [9], where each growth stage couples newly added qubits to a bulk lattice in the way the second term of Eq. (3) couples the appended levels to the coarse space. Ref. [9] turns that bound into explicit sufficient conditions for a Trotterized adiabatic stage: a total time

$$T \;\geq\; \frac{1}{\sqrt{\epsilon}}\int_0^1 \left[\frac{\left\|\ddot{H}\right\|}{\Delta^2} + 7\,\frac{\left\|\dot{H}\right\|^2}{\Delta^3}\right] ds \quad (5)$$

guarantees a final infidelity at most ϵ, with derivatives taken with respect to s and $\Delta(s)$ the instantaneous gap, and $n_s \geq \eta^{-1}(T\Lambda_1/4 + T^2\Lambda_2/2)$ first-order Trotter steps keep the operator error of the digitized evolution below η, with $\Lambda_1 = \max_s \left\|\dot{H}\right\|$ and Λ_2 the largest commutator norm between the groups into which the Hamiltonian is split. Both conditions apply verbatim to Eq. (3), since the path is twice differentiable and the compilation of Sec. V B partitions $H(s)$ into groups of commuting terms; but they are sufficient, not necessary, and Ref. [9] documents for a twelve-site transverse-field Ising chain that they overestimate the time and the step count actually needed by several orders of magnitude — in part because the Trotter error of a digitized adiabatic sweep partly heals itself [11]. The refinement times T^* extracted below are therefore empirical, read off the computed overlap rather than from a bound; Sec. IV E evaluates both conditions for our path and quantifies the overshoot.

D. The rodeo algorithm

The rodeo algorithm [5, 6] removes the residual error left by the adiabatic tail. One cycle acts on the system register plus a single ancilla (reused across cycles by mid-circuit measurement; the circuit is shown in Fig. 11): a Hadamard on the ancilla, the controlled evolution e^{-iHt_k} of the system, a phase gate $P(E\,t_k)$ on the ancilla (which effectively shifts $H \to H - E$), a second Hadamard, and an ancilla measurement. Post-selecting the ancilla in $|0\rangle$ applies the non-unitary filter $\frac{1}{2}\big(\mathbb{1} + e^{iEt_k}e^{-iHt_k}\big)$ to the system: an eigenstate $|E_j\rangle$ acquires the factor $e^{i(E-E_j)t_k/2}\cos[(E_j - E)t_k/2]$. For an eigenstate input the probability that all M ancilla measurements return 0 is therefore [6]

$$P_{0^M}\big(E \mid \{t_k\}\big) = \prod_{k=1}^{M} \cos^2\Big[\tfrac{t_k}{2}\big(E_{\rm obj} - E\big)\Big], \quad (6)$$

and averaging the t_k over a Gaussian of RMS width σ gives

$$P_{0^M}(E) = \left[\frac{1 + e^{-(E_{\rm obj}-E)^2\sigma^2/2}}{2}\right]^M. \quad (7)$$

On resonance ($E = E_{\rm obj}$) the success probability is unity at every cycle, while off-resonant weight is suppressed geometrically toward the background 2^{-M} (the arithmetic mean of $\cos^2$ is 1/2); note that the Gaussian-averaged single-cycle factor $q = \frac{1}{2}[1 + e^{-(E_{\rm obj}-E)^2\sigma^2/2}]$ is never smaller than 1/2, so 2^{-M} is the suppression of components far from resonance, $|E_{\rm obj} - E|\sigma \gg 1$, not a general lower bound on the retained weight. Scanning E thus locates eigenvalues as peaks whose width $\sim 1/\sigma$ sets the energy resolution; fixing E at the target eigenvalue turns the algorithm into a state-preparation filter.

The controlled evolution is the only system-size-dependent ingredient. On hardware e^{-iHt} is realized by a Suzuki–Trotter decomposition over the $2k$ qubits [18–20]; for the single-qubit benchmark of Ref. [6] it reduces to a Z–Y–Z Euler rotation $R_{\hat{n}}(\theta) = e^{-i\theta\,\hat{n}\cdot\vec{\sigma}/2}$ implemented through the Qiskit native gate. In Secs. IV A–IV F we apply the controlled evolution exactly within the conserved N-particle sector to isolate the algorithmic behavior; the circuit-level pipeline with explicit product formulas, and the Trotter error it introduces, are quantified in Sec. V B. Throughout we draw the cycle times t_k from a Gaussian as in Ref. [5]; optimized deterministic [49] or geometric [51] time schedules suppress off-resonant weight faster at fixed runtime and would only improve the numbers reported here, as would the controlled-reversal-gate compilation of the controlled evolution of Ref. [50].

a. Convergence criterion and the random-time ensemble. With E tuned to the target eigenvalue E_0, the all-zero acceptance probability of an input $|\psi\rangle = \sum_k c_k\,|E_k\rangle$ after M cycles, averaged over the Gaussian

time sets, is

$$P_M = p + \sum_{k\neq 0} |c_k|^2 \, q_k^M, \qquad p = |c_0|^2, \tag{8}$$

with $q_k = \frac{1}{2}\big(1 + e^{-(E_k - E_0)^2\sigma^2/2}\big)$ the Gaussian-averaged single-cycle factor of Eq. (7). The target weight p is preserved at every round while off-resonant components decay geometrically, so $P_M \to p$. Two fidelities must be distinguished for a randomized filter. For one fixed time set $\{t_k\}$ the post-selected state has a fidelity $\mathcal{F}(\{t_k\})$ and an acceptance probability $P(\{t_k\})$; the fidelity of the *ensemble of accepted runs*, which is what repeated executions on a device deliver, is the acceptance-weighted average $\mathcal{F}_M = \langle P\,\mathcal{F}\rangle/\langle P\rangle$, and for it the spectral decomposition gives exactly

$$1 - \mathcal{F}_M = \frac{\sum_{k\neq 0} |c_k|^2 q_k^M}{p + \sum_{k\neq 0} |c_k|^2 q_k^M} \leq \frac{(1-p)\, q_{\max}^M}{p + (1-p)\, q_{\max}^M}, \tag{9}$$

with $q_{\max} = \frac{1}{2}\big(1 + e^{-\Delta_{\min}^2\sigma^2/2}\big)$ and $\Delta_{\min}$ the smallest excitation energy carrying weight. The familiar estimate $1 - \mathcal{F}_M \simeq \frac{1-p}{p} 2^{-M}$ is the limit $\Delta_{\min}\sigma \gg 1$ of Eq. (9); for the pairing spectra used below ($\Delta_{\min}\sigma \geq 5$) the two are indistinguishable, and we plot Eq. (9) as the analytic expectation of the ensemble average. The unweighted mean of $\mathcal{F}(\{t_k\})$ over time sets, which some of our figures report, differs from $\mathcal{F}_M$ by less than one percent for inputs with $p > 0.99$ and by a few percent for $p \simeq 0.87$; both are quoted where the difference is visible. Reaching an ensemble fidelity $1-\delta$ needs $M \gtrsim \log_{1/q_{\max}}[(1-p)/(p\,\delta)]$ cycles, and since the expected number of repetitions per accepted run is $1/P_M \to 1/p$, the total controlled-evolution effort scales as $O\big(|\log\delta|/(p\,\epsilon)\big)$ at energy resolution ϵ [5]. The filter is efficient whenever p is close to one, whatever supplies that overlap — resolution refinement, a variational circuit or a cheap correlated reference — which is why the comparison of Sec. V C is between *inputs* to the same filter.

IV. RESULTS

A. Coarse solver: UCCSD versus exact references

Table I lists the ground-state energies at $g = 1$ for $N = 4$ as the basis grows from $k = 2$ to $k = 8$. The gate-level UCCSD circuit reproduces the operator-level (statevector) UCCSD to machine precision and stays variational ($E_{\rm UCCSD} \geq E_{\rm FCI}$), tracking FCI to $\sim 10^{-3}$–10^{-2}, whereas CCD over-binds. Figure 1 compares all solvers (exact, CCD, statevector and gate-level UCCSD, and a hardware-efficient ansatz [14]) as a function of k. The hardware-efficient ansatz is exact at $k = 2$ and reaches FCI at $k = 3$ only after a multi-start optimization, and struggles at $k = 4$ — the symmetry-blind landscape that motivates the physically structured UCCSD circuit. Figure 2 shows that the same UCCSD solver remains a tight variational bound all the way to $k = 8$ (sixteen qubits), at the cost of a CNOT count that grows steeply (1312 CNOTs already at $k = 4$); the compiled circuits and their depths are analyzed in Sec. V. Two remarks keep this table in proportion. First, an energy error is not an infidelity: for a normalized state $\langle H\rangle - E_0 = \sum_{j>0} |c_j|^2 (E_j - E_0) \geq \Delta\,(1 - \mathcal{F})$, so with the physical gap $\Delta = 1.823$ the $k = 4$ energy error 1.44×10^{-3} already caps the infidelity at 7.9×10^{-4}; the actual value, obtained from the optimized state, is $1 - \mathcal{F}_{\rm UCCSD} = 1.78 \times 10^{-4}$, a factor eight below the energy error, and it is this infidelity that every comparison below uses. Second, the scan in k at fixed $N = 4$ is a basis-size study — the fixed-N sector grows as k^4 and the seniority-zero sector as k^2 — and it concerns the UCCSD solver alone, not the complete pipeline; it is not evidence of favorable many-particle scaling at fixed filling.

TABLE I. Ground-state energies at $g = 1$, $N = 4$, for the constant-pairing Hamiltonian. FCI is the exact N-sector result; CCD is coupled-cluster doubles; UCCSD is the gate-level circuit of Sec. III A. "#CNOT" is the two-qubit gate count of the UCCSD state preparation.

k	qubits	FCI	CCD	UCCSD	#CNOT
2	4	1.000000	1.000000	1.000000	0
3	6	0.794697	0.794697	0.794697	272
4	8	0.635548	0.630443	0.636987	1312
6	12	0.372664	0.350058	0.378719	–
8	16	0.145559	0.097263	0.157856	–

B. Resolution refinement: overlap and energy

At $N = 4$ the low-resolution ($k = 2$) space is a single filled determinant, so the prolonged state is the high-resolution HF reference, with first-guess overlaps 0.917 for refinement to $k = 3$ and 0.868 to $k = 4$. Adiabatic evolution along the gapped path of Eq. (3) drives both quantities toward their exact targets: the overlap rises to $\simeq 0.999$ and the energy falls from $E_{\rm HF} = 1$ to the FCI value from above — 0.7964 for $k = 3$ (FCI 0.794697) and 0.6397 for $k = 4$ (FCI 0.635548) at $T = 30$ (Fig. 4). The instantaneous spectrum stays open along the whole path (Fig. 3, right), a necessary condition for the favorable $T \sim 1/\Delta E$ timescale discussed in Sec. III C; the larger $k = 4$ space starts lower in overlap and converges more slowly, as expected from the system-size dependence of the refinement.

C. A nontrivial coarse stage: the $k = 3 \to 4$ chain

At $N = 4$ the $k = 2$ sector is a single determinant, so in the chain above the coarse "solver" is gate-free and the pipeline's first stage does no work. To exer-

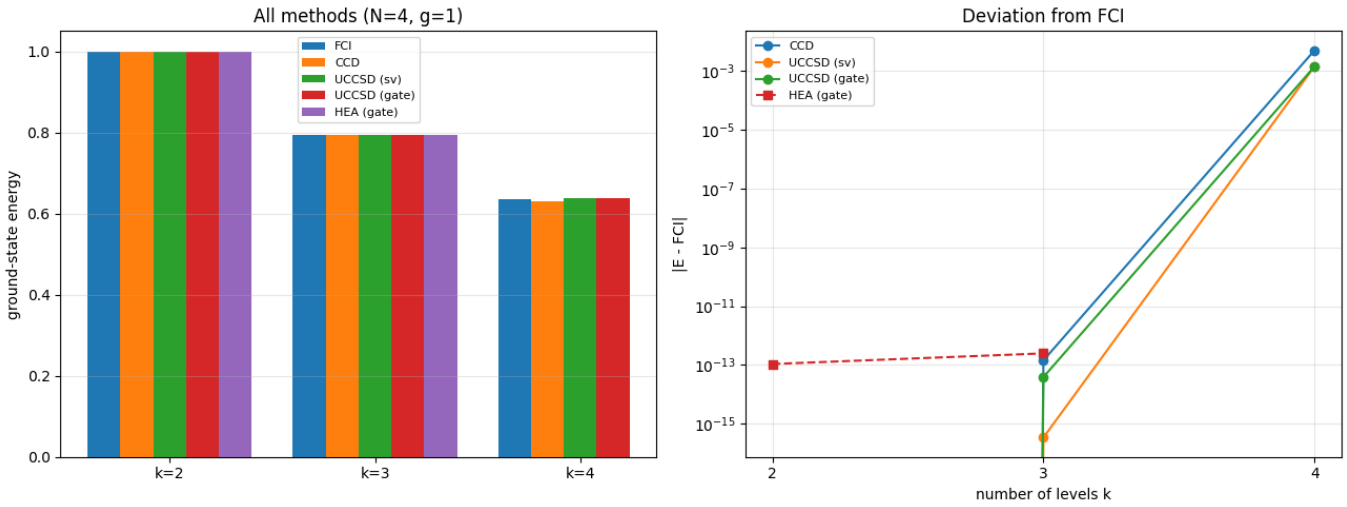


FIG. 1. Ground-state energy at $N = 4$, $g = 1$ obtained with all solvers (left) and their deviation from FCI (right), versus the number of levels k. The structured UCCSD circuits track FCI from above; CCD over-binds; the hardware-efficient ansatz matches FCI for $k \leq 3$ but loses accuracy at $k = 4$.

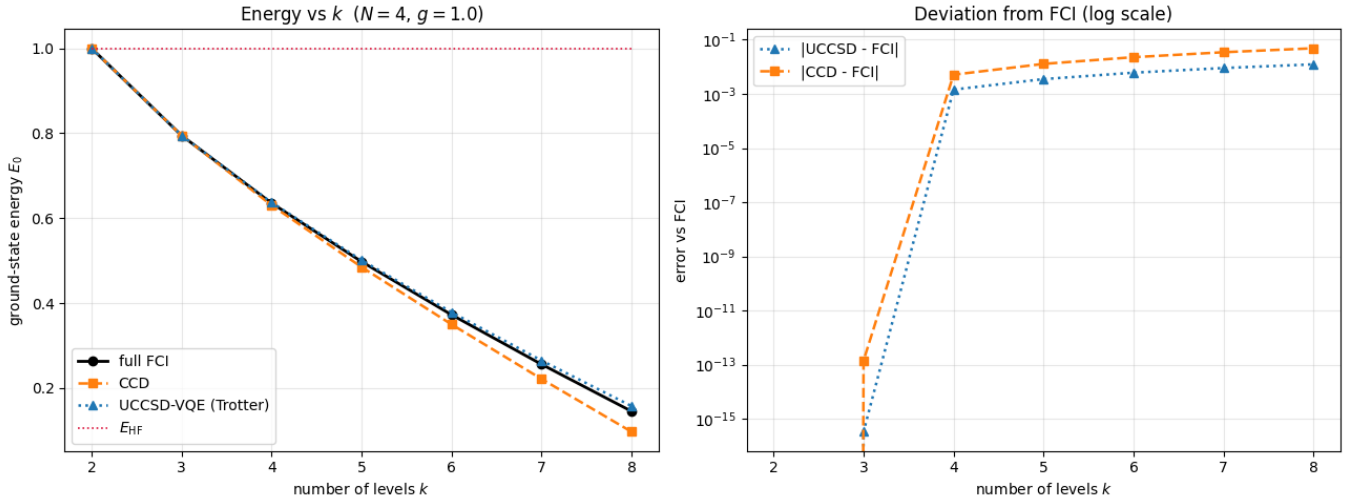


FIG. 2. Scalability of the Trotterized UCCSD–VQE: ground-state energy (left) and deviation from FCI (right) from $k = 2$ to $k = 8$ ($N = 4$, $g = 1$). The same solver remains variational and tight as the basis — and hence the qubit count — grows.

cise the pipeline with a working coarse stage we start instead from the gate-level UCCSD state at $k = 3$ — a genuine 272-CNOT variational preparation, indistinguishable from FCI at this resolution (Table I) — prolong $3 \to 4$, and refine along the same path, with the shift μ now referenced to the *variational* coarse energy $\langle\psi_{\rm low}|H_{\rm low}|\psi_{\rm low}\rangle$, the only quantity a device can know. Figure 5 compares the two chains. The better coarse start raises the prolonged overlap from 0.868 to 0.964 and reaches the target overlap 0.99 at $T^* = 9$ instead of $T^* = 16$. Whether this saves gates is decided by the embedded coarse term of Eq. (4), which is not free when the coarse space is nontrivial. For the $2 \to 4$ chain A is a single four-qubit conditional phase (14 CNOTs per application, Sec. V B); for the $3 \to 4$ chain it is the complete $k = 3$ Hamiltonian conditioned on the two appended qubits being empty, about 350 CNOTs per application, more than the 296 CNOTs of the high-resolution pairing step itself. At the fixed step density of Table III (80 second-order steps per $T = 30$) the $2 \to 4$ chain needs 43 steps for $T^* = 16$, i.e. $2 \times 43 \times (296 + 14) \simeq 2.7 \times 10^4$ CNOTs, and the $3 \to 4$ chain 24 steps for $T^* = 9$, i.e. $2 \times 24 \times (296 + 350) \simeq 3.1 \times 10^4$ CNOTs plus the 272 of the coarse circuit. The shorter adiabatic time of the better start is thus offset by the cost of carrying the coarse Hamiltonian along the path, and the two chains cost the same within 15%. Starting higher in the hierarchy remains the right choice when the prolonged overlap of the trivial start is poor (the strong-coupling regime of

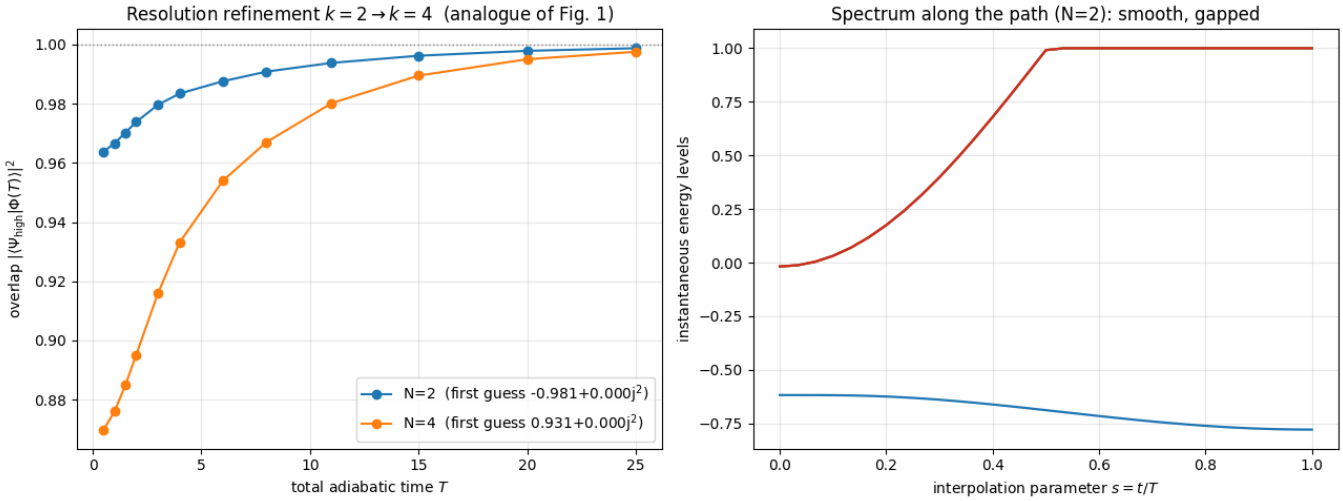


FIG. 3. Resolution refinement $k = 2 \to k = 4$. Left: overlap of the evolved state with the exact high-resolution ground state versus the total adiabatic time T, for $N = 2$ and $N = 4$. Right: the instantaneous spectrum along the interpolation parameter $s = t/T$ stays smooth and gapped, the condition for efficient refinement.

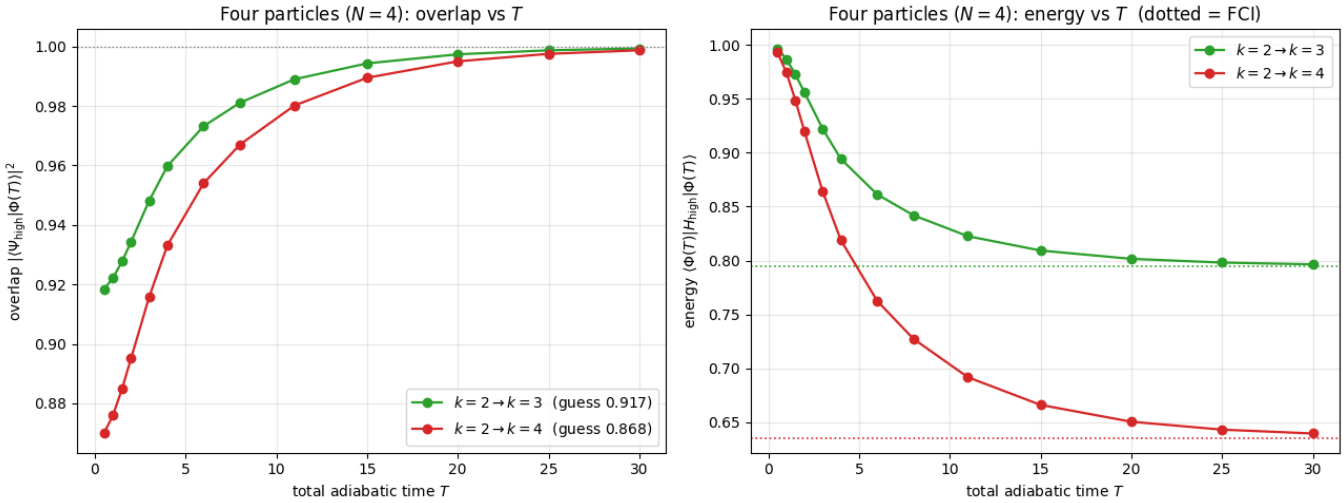


FIG. 4. Four-particle refinement up to $k = 4$: overlap (left) and physical energy $\langle\Phi(T)|H_{\rm high}|\Phi(T)\rangle$ (right) versus T for $k_{\rm high} = 3, 4$. The energy approaches the FCI value (dotted) from above while the overlap approaches unity, both on a $\sim 1/\Delta E$ timescale.

Sec. IV D), but it is not a gate saving in itself.

D. Coupling scan and the path gap

All results above sit at $g = 1$; Fig. 6 probes the refinement across couplings and tests the $T \sim 1/\Delta E$ claim directly. Scanning $g = 0.3$–4 at fixed $\mu = E_{\rm low} + 0.6$ (left panel), the minimum instantaneous gap along the path stays open and nearly constant — it is set by the shift itself, $\Delta E_{\rm min} \simeq \mu - E_{\rm low}$, dropping only to 0.57 at $g = 4$ — while the time to reach overlap 0.99 grows from $T^* = 1$ at $g = 0.3$ to $T^* = 65$ at $g = 4$. The growth tracks the initial overlap of the prolonged state, which falls from 0.99 to 0.38 over the same range: at fixed gap, a poorer starting state simply has more rearrangement to push through the same bottleneck. This identifies the practical limiter of the method at strong coupling — not a closing gap but a degrading prolongation — and it is precisely the regime where a coarse start higher in the hierarchy (Sec. IV C) pays off most.

Because the path gap is controlled by the shift, varying μ at fixed $g = 1$ changes the minimum gap over a factor five (right panel): the extracted T^* is well described by a line in the inverse minimum gap, $T^* \approx 17/\Delta E_{\rm min}$, which is the behavior expected of a first-order-in-$1/\Delta E$ refinement path and incompatible with a $1/\Delta E^2_{\rm min}$ law over this range. The scan does not, however, isolate the

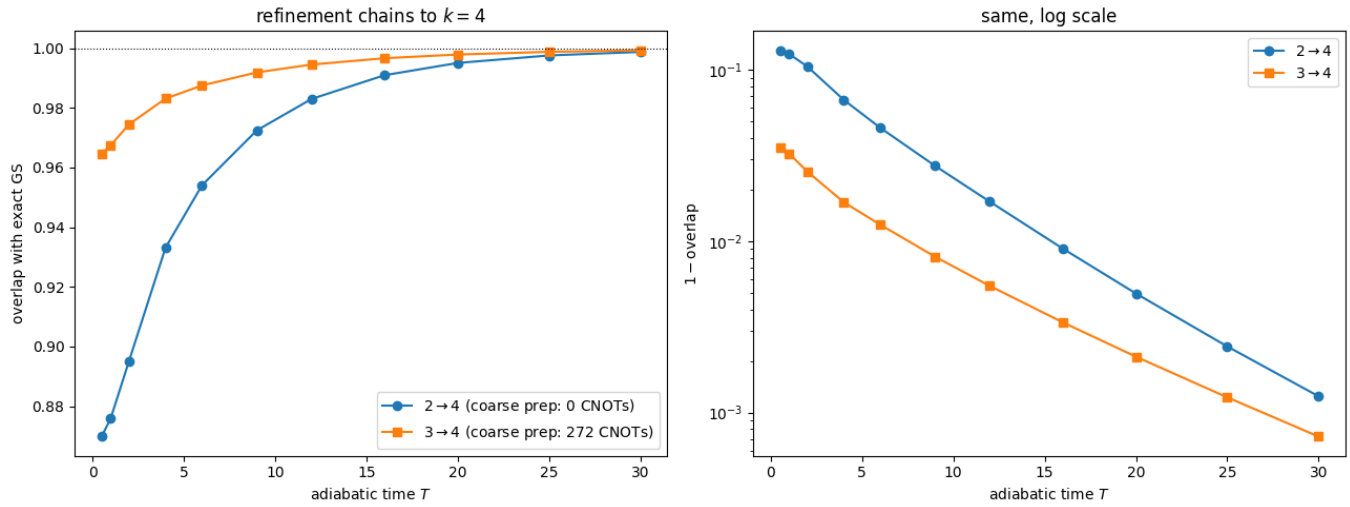


FIG. 5. Refinement chains to $k = 4$ at $N = 4$, $g = 1$, starting from the gate-free $k = 2$ determinant (circles) and from the 272-CNOT gate-level UCCSD state at $k = 3$ (squares). Left: overlap with the exact ground state versus adiabatic time T; right: the same on a logarithmic scale. The $3 \to 4$ chain starts at overlap 0.964 (versus 0.868) and reaches 0.99 in roughly half the adiabatic time.

gap dependence in a strict sense: changing μ changes the interpolation operator $H(s)$ itself, not only its gap, so the result is an empirical trend for this family of paths in a finite system, not a test of the general assumptions under which Ref. [7] derives the inverse-gap scaling. Two gaps should also be kept apart: the shift-controlled gap at $s = 0$, $\mu - E_{\text{low}}$, which is an artificial parameter, and the physical gap of H_{high} at $s = 1$ (1.823 at $g = 1$), which the shift does not touch. The same scan doubles as the sensitivity study for the shift: the choice $\mu_{\text{buf}} = 0.6$ used throughout is conservative, and enlarging the buffer to 1.5 *shortens* the refinement to $T^* = 6$. For a one-dimensional coarse space there is no upper limit to the buffer other than the growth of the energy scale (and hence of the Trotter error) of the path; for coarse spaces with several states the useful initial gap saturates at the coarse excitation gap, as noted in Sec. III C.

E. Sufficient conditions versus practice

The conditions quoted in Sec. III C can be evaluated exactly for the path of Eq. (3). Writing $A = P(H_{\text{low}} - \mu)P^\dagger$ and $B = H_{\text{high}} - \mu$, the schedule gives $\dot{H} = \frac{\pi}{2}\sin(\pi s)\,(B - A)$ and $\ddot{H} = \frac{\pi^2}{2}\cos(\pi s)\,(B - A)$, so the integrand of Eq. (5) is fixed by the single number $\|B - A\|$ and by the instantaneous gap $\Delta(s)$, both taken in the $N = 4$ sector in which the evolution takes place. Figure 7 shows the two along the $k = 2 \to 4$ path: the gap rises from the shift-controlled 0.600 at $s = 0$ to the physical gap of H_{high} at $s = 1$, and the integrand peaks near $s \simeq 0.2$, where $\sin(\pi s)$ is already sizable while the gap is still close to its minimum; the term $7\left\|\dot{H}\right\|^2/\Delta^3$ supplies 94% of the integral. Table II collects the numbers. For the pure pairing model, $\|B - A\| = 7.62$ and $\int_0^1 [\cdots]\, ds = 511$, so Eq. (5) demands $T \geq 5.1 \times 10^3$ for an infidelity of 10^{-2} (and 1.6×10^4 for 10^{-3}), whereas the computed overlap reaches 0.99 at $T^* = 16$: the sufficient condition overshoots the practical time by a factor 3×10^2. The step-count condition is looser still. With $\Lambda_1 = \max_s \left\|\dot{H}\right\| = 12.0$ and, for the term partition of the gate compilation of Sec. V B (the conditioned coarse term, the diagonal term and the six pair-hopping terms), $\Lambda_2 = \max_s \sum_{i<j} \|[H_i(s), H_j(s)]\| = 13.0$, the first-order bound at $\eta = \sqrt{\epsilon}/2$ requires 3.4×10^4 steps along the $T^* = 16$ path — and 3.4×10^9 along the path of length T_{b} — while 160 first-order or 40 second-order steps already reproduce the exact-slice overlap at T^* to within the adiabatic error of the path itself. The rows with $f \neq 0$ anticipate the particle-hole extension of Sec. VI: the time bound hardly moves (4.4–5.1×10^3), because the minimum gap is set by the shift and $\|B - A\|$ changes little, while T^* doubles, so the overshoot falls to 1.4×10^2 at $f = 0.5$; the commutator sum Λ_2, by contrast, grows from 13 to 184 with the additional Trotter groups of V_{ph}, and the step bound at T^* rises to 1.9×10^6 against 640 first-order steps in practice.

These ratios — a few hundred for the time and 2×10^2 to 3×10^3 for the step count — are the fermionic counterpart of what Ref. [9] reports for nucleation of a twelve-site Ising chain (6×10^1 and 10^7, respectively), and they carry the same message: the conditions certify convergence but do not predict cost. The reason is visible in the integrand. The bound pays for the full operator norm $\|B - A\|$, which is dominated by the highest-lying states of H_{high} that the tracked state never couples to, whereas the actual adiabatic error is governed by the matrix elements of dH/ds between the tracked state and

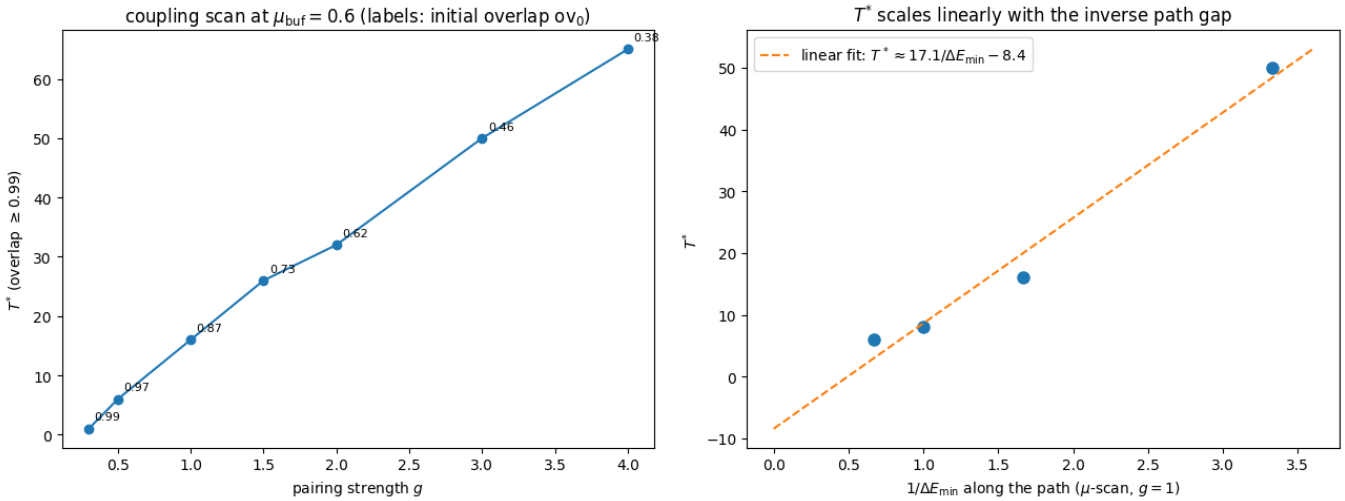


FIG. 6. Left: adiabatic time T^* to reach overlap 0.99 for the $k = 2 \to 4$ refinement versus pairing strength g (point labels: initial overlap of the prolonged state). The minimum path gap stays $\simeq 0.6$ throughout; T^* grows because the starting overlap degrades. Right: T^* versus the inverse minimum path gap, varied through the shift buffer $\mu_{\rm buf} = 0.3$–1.5 at $g = 1$; the linear fit confirms $T^* \sim 1/\Delta E_{\rm min}$.

the few states it does couple to, relative to their gaps — precisely the quantity whose boundedness underlies the inverse-gap scaling of Ref. [7] and the linear trend of Fig. 6.

F. Rodeo filtering of the refined state

We verified the rodeo primitive in two ways: the explicit ancilla circuit agrees with the compact filter $\frac{1}{2}(\mathbb{1} + e^{iEt}e^{-iHt})$ to 10^{-16}, and the all-zero probabilities reproduce Eqs. (6)–(7), with the Gaussian-averaged single-cycle factor matching $\frac{1}{2}(1 + e^{-(\Delta E\,\sigma)^2/2})$ to three digits.

The state fed to the filter is the resolution-refined $k = 4$ state of Sec. IV B at $T = 25$, with overlap $p = 0.9976$ with the exact ground state; for contrast we also use the unrefined prolonged guess Φ_0 ($p = 0.868$). Scanning the target energy E (Fig. 8; $M = 4$ cycles, 12 time sets per energy) produces a peak at the ground-state energy; a coarse scan ($\sigma = 2$) resolves it from the excited band, and a fine scan ($\sigma = 8$) locates it at $E = 0.6352$ against the exact $E_0 = 0.635548$. Starting from the refined state, essentially all spectral weight sits in the single ground-state peak; starting from Φ_0, the small residual weight on excited states is visible as secondary structure above the 2^{-M} background. The scan is part of the cost of the method: the fine scan alone evaluates 161 energies with 12 time sets each, i.e. about 2×10^3 four-cycle runs, each repeated over enough shots to estimate an acceptance probability.

Fixing $E = E_0$ and applying M cycles with $\sigma = 3$ (Fig. 9; 40 Gaussian time sets, unweighted mean of the post-selected fidelity), the fidelity rises geometrically and the cumulative acceptance settles at $\approx p$, in quantitative agreement with Eqs. (8)–(9). The refined input reaches an infidelity $1-\mathcal{F} = 1.4\times10^{-3}$ after one cycle, 6.5×10^{-4} after two, 2.1×10^{-5} after six and 2×10^{-6} after ten; the prolonged input reaches 1.3×10^{-4} after ten cycles while its acceptance holds at 0.868. The dotted lines are the analytic ensemble expectation of Eq. (9); the sampled points scatter around them by up to ±30% at the smallest infidelities, the statistical precision of 40 time sets (the acceptance-weighted ensemble fidelity agrees with Eq. (9) to better than 1% for the refined input). Each cycle buys roughly a factor two in infidelity, so reaching 10^{-6} from $p = 0.998$ takes ten cycles, not one or two. Two practical checks complete the picture. First, the target energy on a device is the scan estimate, not the exact eigenvalue: repeating the filter at $E = 0.6352$ instead of E_0 changes no digit of the infidelities quoted above. Second, a deliberately detuned target attenuates the ground state as well: for $E - E_0 = 0.05$, 0.1 and 0.2 the six-cycle infidelity is unchanged within 10%, 35% and a factor 2.7, while the acceptance falls from 0.998 to 0.971, 0.895 and 0.656 — the acceptance, not the fidelity, is the sensitive diagnostic of a wrong target, and it then no longer tends to the input overlap.

V. QUANTUM CIRCUITS AND DEPTH COMPARISON

All quantum operations in this work are compiled to a common gate set of single-qubit rotations and CNOTs, so the resource costs of the different routes to the high-resolution ground state — a full-space UCCSD–VQE, the refinement-plus-filter pipeline, and the filter applied directly to cheaper inputs — can be compared gate by gate.

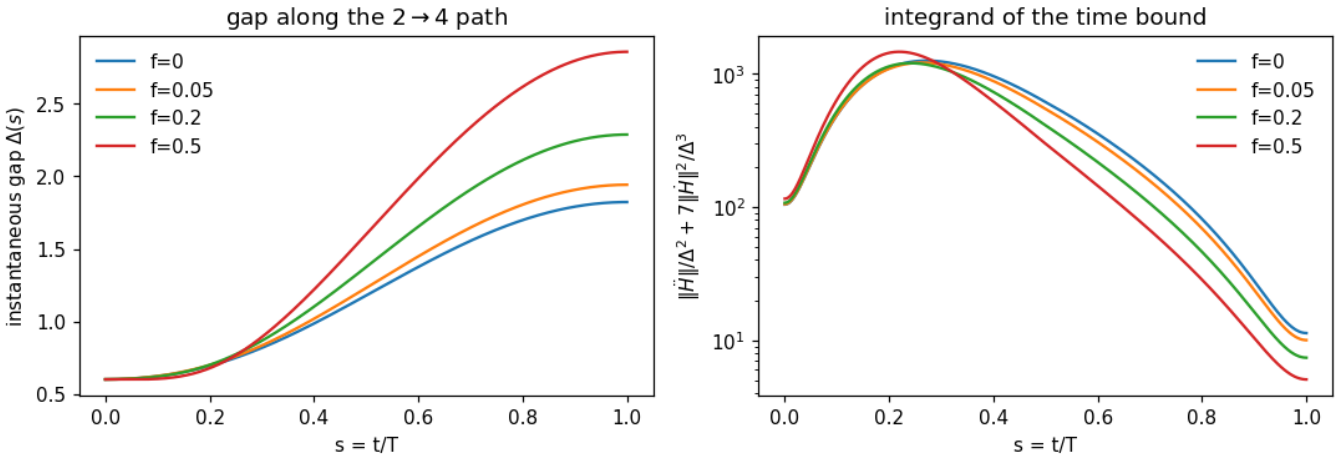


FIG. 7. The ingredients of the adiabatic-time bound, Eq. (5), along the $k = 2 \to 4$ refinement path at $N = 4$, $g = 1$, $\mu = E_{\rm low} + 0.6$, for the pure pairing model and the three particle-hole strengths of Sec. VI. Left: instantaneous gap $\Delta(s)$ in the $N = 4$ sector, from the shift-controlled 0.600 at $s = 0$ to the physical gap of $H_{\rm high}$ at $s = 1$. Right: the integrand $\left\|\ddot{H}\right\|/\Delta^2 + 7\left\|\dot{H}\right\|^2/\Delta^3$ (logarithmic scale), which peaks near $s \simeq 0.2$ and integrates to 435–511 (Table II).

TABLE II. Sufficient conditions of Ref. [9] evaluated for the $k = 2 \to 4$ refinement path ($N = 4$, $g = 1$, $\mu = E_{\rm low} + 0.6$; spectral norms in the $N = 4$ sector) against the empirical cost. $\Delta_{\rm min}$: minimum instantaneous gap; I: the integral in Eq. (5); $T_{\rm b}$: the time bound for infidelity $\epsilon = 10^{-2}$, $T_{\rm b} = I/\sqrt{\epsilon}$; T^*: time for overlap 0.99 (grid of Fig. 6); $\Lambda_1 = \max_s \left\|\dot{H}\right\|$ and $\Lambda_2 = \max_s \sum_{i<j} \|[H_i, H_j]\|$ over the Trotter groups of the gate compilation; $n_{\rm b}(T^*)$: first-order Trotter steps required by the bound along the T^* path at $\eta = \sqrt{\epsilon}/2$; n_1, n_2: first- and second-order steps (from the sequence $5, 10, 20, \ldots$) that reproduce the exact-slice overlap at T^* to within the adiabatic error $1 - p(T^*)$ of the path. The rows with $f \neq 0$ refer to the particle-hole model of Sec. VI.

f	$\Delta_{\rm min}$	$\|B - A\|$	I	$T_{\rm b}$	T^*	$T_{\rm b}/T^*$	Λ_1	Λ_2	$n_{\rm b}(T^*)$	n_1	n_2
0	0.600	7.62	511	5.1×10^3	16	3.2×10^2	12.0	13.0	3.4×10^4	160	40
0.05	0.600	7.66	480	4.8×10^3	20	2.4×10^2	12.0	21.6	8.8×10^4	320	40
0.2	0.600	7.85	435	4.3×10^3	26	1.7×10^2	12.3	58.7	4.0×10^5	320	80
0.5	0.600	8.42	446	4.5×10^3	32	1.4×10^2	13.2	184	1.9×10^6	640	160

Every number quoted below is produced in the accompanying notebook. Two levels of simulation must be distinguished. The Pauli-rotation primitive, the UCCSD excitation circuits, the hardware-efficient ansatz and the ancilla-based rodeo cycle are simulated *gate by gate* on the full $2k$-qubit register and verified against exact exponentials. The product-formula evolutions of Sec. V B, by contrast, are simulated *block by block*: each term of the Hamiltonian splitting is a set of mutually commuting Pauli strings whose exponential is realized exactly by the corresponding rotation–CNOT block, and we apply that exact block exponential within the conserved N-particle sector. This reproduces faithfully the state that the compiled circuit would produce, and the CNOT counts are those of the compiled blocks, but it does not execute every individual rotation and CNOT; we therefore quote CNOT *counts* rather than scheduled circuit depths, which depend on the connectivity and parallelism of a device.

A. Circuit building blocks

The elementary primitive is the Pauli rotation $e^{-i\frac{\phi}{2}P}$ for a Pauli string $P = \bigotimes_q \sigma_q$ [37]: each active qubit is rotated into the Z eigenbasis (H for X, $HS^\dagger$ for Y), a CNOT ladder accumulates the parity onto the last active qubit, a single $R_z(\phi)$ is applied, and the ladder and basis changes are uncomputed. A string of weight w therefore costs $2(w-1)$ CNOTs and one parametrized rotation; the circuit reproduces the exact exponential to 3×10^{-16}.

a. UCCSD preparation circuit. Through the Jordan–Wigner map a single excitation compiles into 2 Pauli rotations whose weight grows with the distance between the spin orbitals (8 CNOTs for adjacent levels, 16 for the $0 \to 4$ single at $k = 3$), and a generic double excitation into 8 rotations (48–64 CNOTs). The doubles that dominate the pairing problem are special: for a time-reversed pair moving between levels p and q the operators act on the adjacent qubits $(2p, 2p{+}1)$ and

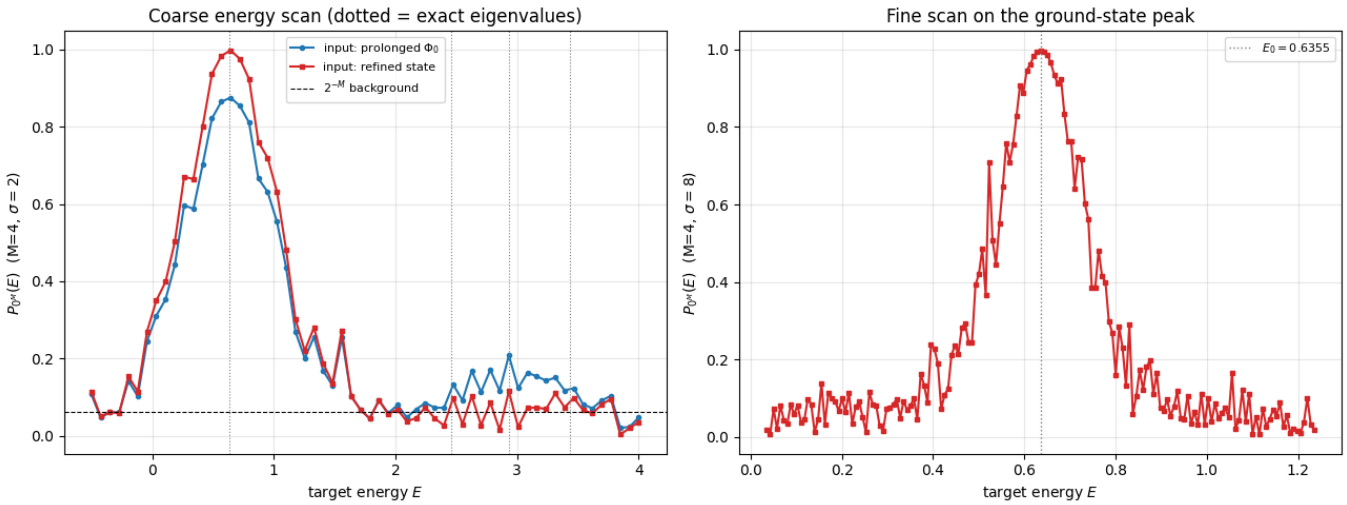


FIG. 8. Rodeo energy scan ($M = 4$). Left: coarse scan ($\sigma = 2$); the dotted vertical lines mark exact eigenvalues and the dashed line the 2^{-M} background. Right: fine scan ($\sigma = 8$) on the ground-state peak, located at $E = 0.6352$ (exact $E_0 = 0.6355$).

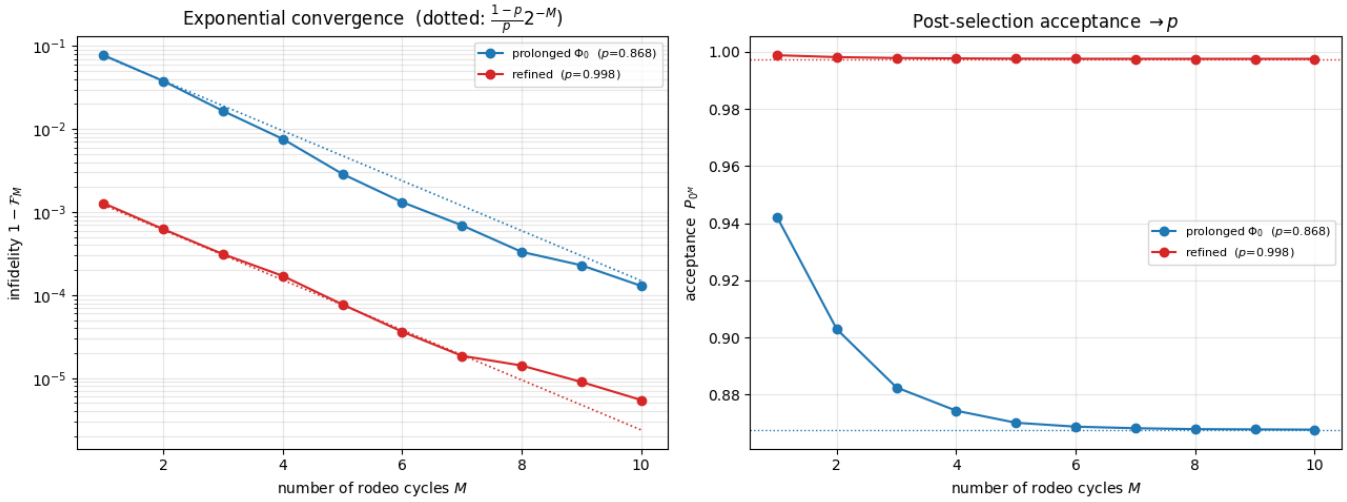


FIG. 9. Rodeo eigenstate preparation at $E = E_0$ ($k = 4$, $N = 4$, $g = 1$, $\sigma = 3$), averaged over 40 Gaussian time sets, for the resolution-refined input ($T = 25$, exact slices, $p = 0.9976$) and the prolonged determinant ($p = 0.868$), with exact controlled evolution. Left: unweighted mean of the post-selected infidelity versus the number of cycles M (dotted: the analytic ensemble expectation of Eq. (9), which for these spectra coincides with $\frac{1-p}{p}2^{-M}$). Right: the post-selection acceptance converges to the input overlap p.

$(2q, 2q{+}1)$, so the internal Jordan–Wigner Z strings cancel and all 8 Pauli strings have weight 4 — 48 CNOTs per pair-hopping term *independent of the level separation*. The full state preparation at $N = 4$, $g = 1$ then costs 8 excitation operators, 40 rotations and 272 CNOTs at $k = 3$, and 26 operators, 160 rotations and 1312 CNOTs at $k = 4$ (Table I); at $k = 2$ the ansatz is empty (the sector is a single determinant) and the preparation is gate-free.

Figure 10 shows both levels of this hierarchy. Panel (a) is the VQE at the ansatz level: the product of excitation unitaries acts on the HF register, the energy is estimated from repeated measurements, and a classical optimizer updates the P amplitudes $\boldsymbol{\theta}$ before the circuit is run again — it is this feedback loop, wrapped around the full-depth state preparation, that multiplies the 1312-CNOT cost into the total gate budget of the traditional route. Panel (b) is the compiled interior of a single weight-4 Pauli rotation, the sub-block that occurs eight times in every pair-hopping double and 160 times in the $k = 4$ preparation; its only parametrized element is the central $R_z(\phi)$, with ϕ fixed by the amplitude of the parent excitation operator.

b. Hardware-efficient ansatz. The HEA [14] used for contrast consists of L layers of R_y rotations on all $2k$ qubits interleaved with a CNOT entangler (nearest-

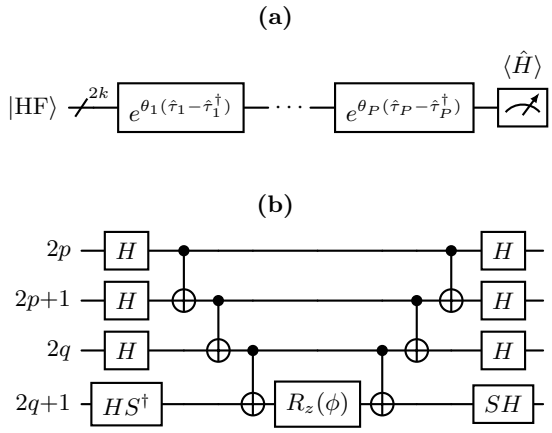


FIG. 10. The variational quantum eigensolver as a circuit. (a) Ansatz level: the Trotterized UCCSD product acts on the HF register ($2k$ qubits, slashed wire) and the energy is estimated by measuring $\langle \hat{H} \rangle$ (in practice a sum of Pauli expectation values, each averaged over many shots); a classical optimizer updates $\boldsymbol{\theta}$ and the entire circuit is re-executed, once per energy or gradient evaluation. (b) Compiled circuit of one weight-4 Pauli rotation, here $e^{-i\frac{\phi}{2}X\otimes X\otimes X\otimes Y}$ acting on the four qubits of a pair-hopping double excitation $(2p, 2p{+}1) \to (2q, 2q{+}1)$: basis changes into the Z eigenbasis (H for X, $HS^\dagger$ for Y), a CNOT parity ladder, the single parametrized $R_z(\phi)$, and the uncompute. Eight such blocks (48 CNOTs) build one double excitation; 160 of them (1312 CNOTs) build the $k = 4$ preparation of Table I.

neighbor pairs plus a wrap-around bond), i.e. $2k(L{+}1)$ variational angles and only $O(2kL)$ CNOTs: 12 angles at $k = 2$ ($L = 2$) and 30 at $k = 3$ ($L = 4$). Its depth is far smaller than UCCSD, but the circuit carries no particle-number or seniority structure; a number-penalty term and multi-start optimization are needed at $k = 3$, and at $k = 4$ the optimizer fails to reach the ground state (Fig. 1). Depth is not the only currency: trainability of the symmetry-blind landscape becomes the bottleneck first.

c. Rodeo cycle. One rodeo cycle (Fig. 11) adds a Hadamard on the ancilla, the controlled evolution e^{-iHt_k}, the ancilla phase $P(Et_k)$ — the element that effectively shifts $H \to H - E$ — a second Hadamard, and a mid-circuit measurement; the single ancilla is reset and reused across the M cycles. The contrast with Fig. 10(a) is the structural point of this section: the rodeo circuit contains *no* variational parameters and closes no optimization loop. The times t_k are drawn once from a Gaussian, the target energy E is a classical setting, and the algorithm is driven entirely by the measurement record — a run is accepted when all M ancilla outcomes read 0, which applies the filter of Sec. III D cycle by cycle. Where the VQE re-executes its full-depth circuit to steer $\boldsymbol{\theta}$, the rodeo circuit is executed as printed, and its only system-size-dependent element is the controlled evolution. The controlled evolution is compiled from the same Pauli-rotation primitive: in each rotation sub-circuit only the central R_z needs to be controlled (if the rotation is absent the basis changes and CNOT ladder cancel), and a controlled R_z costs 2 extra CNOTs. For the pairing Hamiltonian (1) a first-order Trotter step contains the $k(k{-}1)/2$ pair-hopping terms (8 weight-4 rotations, 48 CNOTs each), k diagonal ZZ rotations (2 CNOTs each)

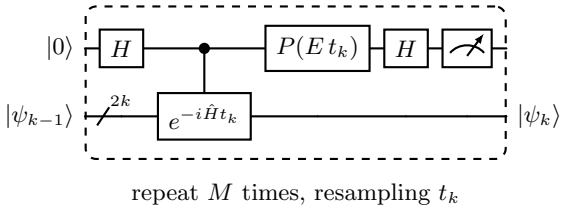


FIG. 11. One cycle of the rodeo algorithm: Hadamard on the ancilla, controlled time evolution $e^{-i\hat{H}t_k}$ of the $2k$-qubit system register with a Gaussian-random time t_k, ancilla phase $P(Et_k)$, second Hadamard, and a mid-circuit measurement of the ancilla, which is then reset and reused for the next cycle. A run is accepted when all M outcomes read 0; each accepted cycle applies the filter $\frac{1}{2}(\mathbb{1} + e^{iEt_k}e^{-i\hat{H}t_k})$ to the system. Unlike the VQE of Fig. 10(a), the circuit carries no variational parameters and no classical optimization loop; the controlled evolution, compiled from the same Pauli-rotation blocks as Fig. 10(b) with only the central R_z controlled, is the sole system-size-dependent element ($\approx 4.2 \times 10^2$ CNOTs per Trotter step at $k = 4$).

and $2k$ single-qubit phases, i.e.

$$N_{\mathrm{CNOT}}^{\mathrm{step}}(k) = 24\,k(k-1) + 2k \tag{10}$$

CNOTs uncontrolled — 296 at $k = 4$ — and $\approx 4.2 \times 10^2$ CNOTs per step once the controls are added. A rodeo cycle with n_T Trotter steps therefore costs $\approx 4.2 \times 10^2\, n_T$ CNOTs at $k = 4$.

B. The Trotterized pipeline

The results of Sec. IV F used exact sector exponentials; on hardware both the refinement slices and the controlled

evolution are product formulas, and their error must be quantified at the circuit level. We split the Hamiltonian exactly as the gate compilation does: a diagonal term D (kinetic energy plus the $p{=}q$ pairing term; single-qubit phases and ZZ rotations) and one pair-hopping term h_{pq} per level pair. The eight Pauli strings of each h_{pq} commute, so every factor $e^{-ih_{pq}\delta t}$ is realized *exactly* by its 48-CNOT block, and the only Trotter error is the non-commutativity between terms — exactly as on a device. At $k = 4$, $t = 1$ the second-order (symmetric) formula reaches spectral-norm error 3.4×10^{-2} with 4 steps and 8.3×10^{-3} with 8.

a. The embedded coarse term. The refinement path, Eq. (3), contains a third kind of factor, $e^{-i\cos^2(\pi s/2)A\,\delta t}$ with $A = P(H_{\rm low} - \mu)P^\dagger$ of Eq. (4). Gate-free prolongation does not make this evolution gate-free. For the $2 \to 4$ chain at $N = 4$ the coarse space is the single determinant $|\mathrm{HF}\rangle$, so $A = (E_{\rm low}{-}\mu)\,|\mathrm{HF}\rangle\langle\mathrm{HF}| = -0.6\;|\mathrm{HF}\rangle\langle\mathrm{HF}|$, and within the $N = 4$ sector $|\mathrm{HF}\rangle$ is the unique state with the four appended qubits 4–7 all in $|0\rangle$. Its exponential is therefore a phase conditioned on four qubits being empty, $X^{\otimes 4}\,C^3P(\phi)\,X^{\otimes 4}$ with $\phi = 0.6\cos^2(\pi s/2)\delta t$. Expanding the projector, $|1111\rangle\langle 1111| = 2^{-4}\sum_S (-1)^{|S|} Z_S$ over the 15 nonempty subsets S of the four qubits, gives a phase polynomial of 15 commuting Z-strings; ordered along a Gray code it compiles into 14 CNOTs and 15 R_z rotations (the naive ladder-per-string construction costs 34 CNOTs). We verified that this operator reproduces the sector matrix exponential of A exactly. For a chain that starts from a nontrivial coarse space the term is $(H_{\rm low} - \mu)$ on the coarse register conditioned on the appended qubits being empty; for the $3 \to 4$ chain each of the 33 Pauli rotations of a $k = 3$ step acquires a double control (6 CNOTs per rotation with a three-qubit Gray-code phase polynomial), for about 350 CNOTs per application, which is the figure used in Sec. IV C. All refinement budgets below include the coarse term; for the $2 \to 4$ chain it adds 14 CNOTs to the 296 of a first-order step, i.e. $2 \times 80 \times 14 = 2\,240$ CNOTs or 5% over the $T = 30$ path.

Figure 12 shows the two pipeline stages under this compilation. For the refinement (left), $n_s = 80$ second-order steps along the $T = 30$ path reproduce the exact-slice result to within its own adiabatic error, delivering $p = 0.9977$ against the exact-slice 0.9988; first-order steps need several hundred. For the rodeo stage (right) the honest scheme fixes the maximum Trotter step *size* δt, so a cycle with Gaussian time t_k uses $n_k = \lceil |t_k|/\delta t \rceil$ steps of size t_k/n_k ($\sigma\sqrt{2/\pi}/\delta t \approx 13$ steps on average at $\sigma = 4$, $\delta t = 0.25$). The filter is strikingly diagnostic of its own Trotter error: at $\delta t = 1$ the corrupted evolution is simply *rejected* — the acceptance collapses to 0.10 and the fidelity never converges — while at $\delta t = 0.25$ the infidelity tracks the exact-evolution curve down to 1.2×10^{-4} at $M = 6$ with acceptance 0.989. The intermediate $\delta t = 0.5$ converges to a floor of 2×10^{-3}. The picture of Ref. [6] — the filter projects onto the ground state of an *effective* Trotterized Hamiltonian, whose $O(\delta t^2)$ rotation away from the true ground state sets the attainable fidelity — describes this behavior, with one qualification: because the step size t_k/n_k differs from cycle to cycle, the cycles do not share a single effective Hamiltonian, and the observed floor is the result of a mixture of nearby effective ground states rather than a projection on one of them. Coherent product-formula errors can thus both shift the located energy and bias the prepared state; the rejection of grossly inaccurate evolutions seen at $\delta t = 1$ is a useful diagnostic, not a guarantee that smaller coherent errors are removed.

C. Gate budget at matched fidelity: which input should the filter get?

The question that decides the value of the refinement stage is not whether the pipeline beats an *unfiltered* variational state — a few rodeo cycles improve any input with large overlap — but whether refinement followed by the filter beats the same filter applied to the best input that is available cheaply. We therefore feed one and the same Trotterized filter (second order, $\delta t = 0.25$, $\sigma = 4$, $|t_k|$ drawn from a Gaussian, 40 common time sets) four inputs: (i) the prolonged $k = 2$ determinant, $p = 0.868$, gate-free; (ii) the full-space gate-level UCCSD state, $p = 0.99982$, 1 312 CNOTs; (iii) the Trotterized refined state at $T^* = 16$ (43 second-order steps, $p = 0.9922$, $2 \times 43 \times 310 = 26\,660$ CNOTs including the coarse term); and (iv) the refined state at $T = 30$ of Fig. 12 (80 steps, $p = 0.9977$, 49 600 CNOTs). For each we track the ensemble infidelity of the accepted runs, the acceptance P_M, and the CNOTs per accepted preparation,

$$C_M = \frac{C_{\rm prep} + \sum_{m=1}^{M} \langle P_{m-1}\rangle\, c_m}{P_M}, \tag{11}$$

where $c_m = 2 \times 416\, n_m$ is the cost of the m-th controlled cycle and the factor $\langle P_{m-1}\rangle$ accounts for early termination of a run at the first ancilla outcome 1 (without early termination the numerator is simply $C_{\rm prep}{+}\sum_m c_m$; the difference is below 5% for all inputs). Equation (11) counts repeated state preparation only. One-time costs are reported separately: the UCCSD training took eight L-BFGS iterations, each requiring the energy and its 26-component gradient, i.e. of order 30 distinct 1 312-CNOT circuits per iteration times the shots needed for the expectation values; the energy search of the rodeo stage costs the scans of Sec. IV F, of order 10^3 four-cycle runs times their shots, and is common to all inputs.

Figure 13 and Table III give the result. The first observation is that every input converges to the same floor, $1{-}\mathcal{F} \simeq 1.1$–$1.5{\times}10^{-4}$ after six cycles, set by the product-formula evolution at $\delta t = 0.25$ and not by the input. The second is the ranking at fixed cycle number. The UCCSD state, whose actual infidelity 1.78×10^{-4} is already close to that floor, reaches 1.3×10^{-4} after two cycles for 2.3×10^4 CNOTs per accepted preparation; the refined $T = 30$ state reaches 5.3×10^{-4} after two

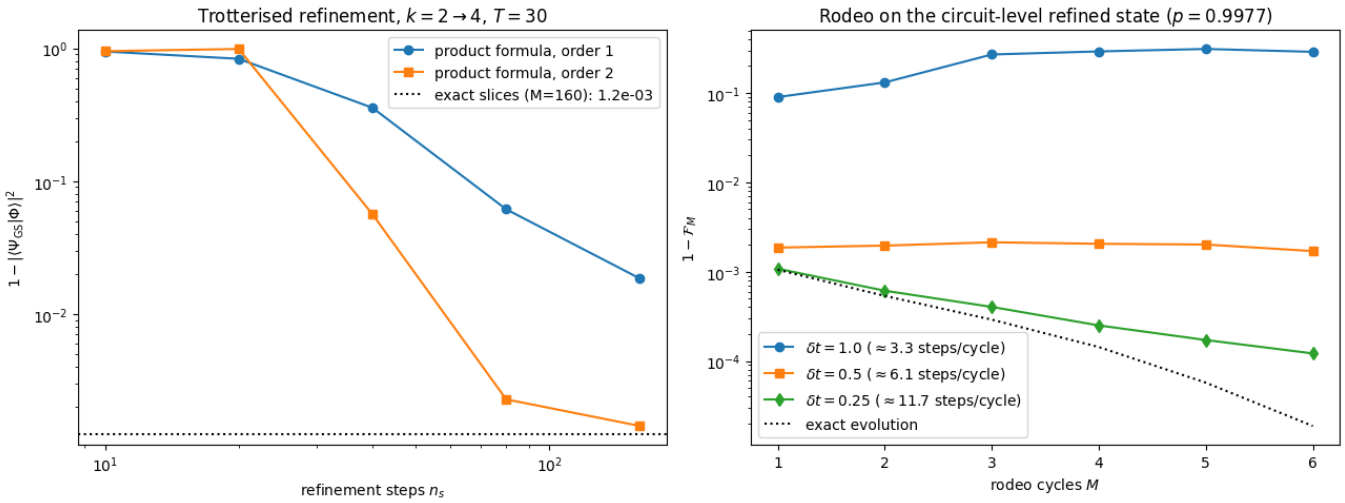


FIG. 12. The pipeline at the circuit level ($k = 4$, $N = 4$, $g = 1$). Left: infidelity of the Trotterized refinement versus the number of product-formula steps n_s along the $T = 30$ path, for first- and second-order formulas; the dotted line is the exact-slice reference. Right: rodeo infidelity versus the number of cycles M for a Trotterized controlled evolution with fixed step size δt (input: the $n_s = 80$ circuit-level refined state, $p = 0.9977$). Too-coarse steps are rejected by the filter itself (the $\delta t = 1$ acceptance collapses to 0.10); $\delta t = 0.25$ tracks the exact evolution.

cycles for 7.2×10^4 CNOTs and 1.5×10^{-4} after six for 1.2×10^5; the refined $T^* = 16$ state is cheaper at every cycle number than the $T = 30$ state at equal or better accuracy beyond $M = 4$, which shows that the refinement time of Table III was not chosen economically; and the unrefined prolonged determinant, filtered directly, reaches 1.8×10^{-3} after six cycles for 7.0×10^4 CNOTs, i.e. roughly the accuracy of the $T^* = 16$ chain after two cycles (1.7×10^{-3}) at roughly 1.4 times its cost. At matched fidelity, therefore, refinement followed by the filter is tied within a factor 1.5 with filtering the unrefined input, and costs three times more than filtering the UCCSD input while delivering a lower fidelity. The claim that the hybrid undercuts the variational preparation by orders of magnitude, which compared the UCCSD *energy error* with the pipeline *infidelity*, does not survive the comparison of like with like.

What remains in favor of the refinement stage is qualitative but real. It contains no variational parameters: nothing is trained at full resolution, no expectation values are measured during state preparation, and the only classical inputs are the shift μ and the schedule T, which the coupling and shift scans of Sec. IV D set from coarse-space information. Its gate count is dominated by the refinement, which scales with the Trotter step of the fine Hamiltonian and with T^*, and is independent of how hard the variational landscape is. Whether this trade — trainability for depth — pays off is a question about regimes in which the full-space variational input is poor or expensive to train. For the pairing model with $N = 4$ that regime is not reached at any size we can validate exactly (Sec. V E), and we state this as the main negative finding of the present assessment.

D. Remarks on adaptive and improved VQE

The steep UCCSD gate count is not irreducible, and much recent work aims precisely at shallower variational circuits (see Ref. [17] for a review). ADAPT-VQE [15] grows the ansatz one operator at a time, at each step appending the pool operator with the largest energy gradient and re-optimizing all amplitudes; it typically reaches a target accuracy with a small fraction of the full UCCSD operator count and correspondingly shallower circuits. The pairing model makes the mechanism transparent: because the Hamiltonian conserves seniority and the reference is seniority-zero, the gradients of all singles and of all pair-breaking doubles vanish identically, so an adaptive selection would retain only the $\frac{N}{2}(k - \frac{N}{2})$ pair-hopping doubles — four operators at $k = 4$, $N = 4$, i.e. $4 \times 48 = 192$ CNOTs in place of the 1312 of the full 26-operator ansatz, a sevenfold compression, with the discarded amplitudes expected to vanish at the optimum just as the singles do. Qubit-ADAPT-VQE [16] pushes further by drawing the pool from individual Pauli strings, removing the eightfold string overhead per fermionic double. These developments are complementary to the present pipeline rather than competing with it: the filter is agnostic to how its input is produced, so an ADAPT-style state — at the coarse or at the full resolution — slots in directly, and in the light of Table III a pair-adapted full-space ansatz of 192 CNOTs followed by two rodeo cycles would be the cheapest route of all at this system size. We have not implemented ADAPT-VQE here; its inclusion as a comparator is the natural next step of the assessment. Conversely, no variational compression by itself supplies the geometric, optimizer-

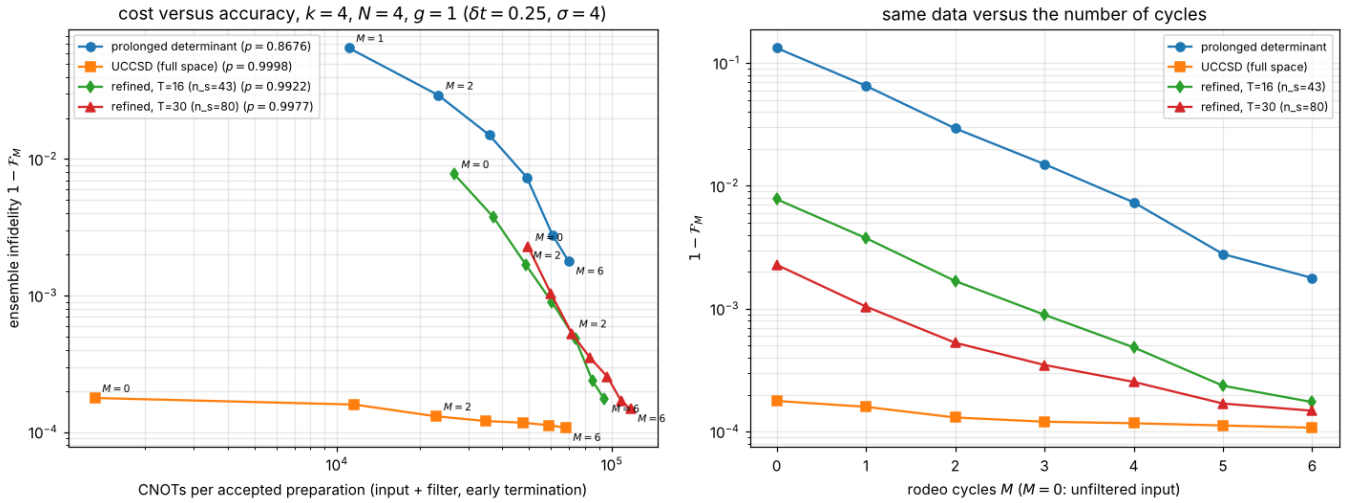


FIG. 13. Matched-observable comparison of inputs to the same Trotterized rodeo filter ($k = 4$, $N = 4$, $g = 1$; second order, $\delta t = 0.25$, $\sigma = 4$, 40 common Gaussian time sets): the prolonged $k = 2$ determinant, the full-space gate-level UCCSD state, and the Trotterized resolution-refined state at $T^* = 16$ and $T = 30$ (the refinement cost includes the coarse term of Eq. (4)). Left: acceptance-weighted ensemble infidelity of the accepted runs versus CNOTs per accepted preparation, Eq. (11), as the number of cycles grows from $M = 0$ (unfiltered input; omitted for the gate-free determinant) to $M = 6$. Right: the same infidelities versus M. All inputs converge to the floor set by the product formula; the UCCSD input reaches it first and cheapest.

TABLE III. Gate budget per accepted preparation at $k = 4$ ($N = 4$, $g = 1$, eight system qubits) for four inputs to the same Trotterized rodeo filter ($\delta t = 0.25$, $\sigma = 4$, 40 time sets). p is the input overlap, $C_{\rm prep}$ the CNOT count of the input (refinement: $2\,n_s$ second-order half-steps of $296 + 14$ CNOTs, the 14 being the conditional phase of the coarse term), $1 - \mathcal{F}_M$ the acceptance-weighted infidelity of the accepted runs, P_M the acceptance and C_M the CNOTs per accepted preparation of Eq. (11). One-time training (UCCSD) and the energy scan are not included (see text). The UCCSD energy error of 1.4×10^{-3} (Table I) corresponds to the infidelity 1.8×10^{-4} listed here.

			$M = 2$			$M = 6$		
input to the filter	p	$C_{\rm prep}$	$1 - \mathcal{F}_2$	P_2	C_2	$1 - \mathcal{F}_6$	P_6	C_6
prolonged determinant (k=2→4)	0.868	0	2.9×10^{-2}	0.887	2.3×10^4	1.8×10^{-3}	0.854	7.0×10^4
UCCSD, full space (26 parameters)	0.99982	1 312	1.3×10^{-4}	0.996	2.3×10^4	1.1×10^{-4}	0.986	6.8×10^4
refined, T^*=16 (n_s=43)	0.9922	26 660	1.7×10^{-3}	0.989	4.9×10^4	1.7×10^{-4}	0.979	9.4×10^4
refined, T=30 (n_s=80)	0.9977	49 600	5.3×10^{-4}	0.994	7.2×10^4	1.5×10^{-4}	0.985	1.2×10^5

free convergence that the rodeo filter provides once the overlap is large. A different route to a shallower circuit — keeping the Trotterized refinement path as the ansatz structure and optimizing a few of its layers, as in the optimized nucleation of Ref. [9] — is taken up in Sec. VII.

E. Other model-space sizes, particle numbers, and observables

To see whether the ranking of Table III is an accident of the smallest nontrivial case, Table IV repeats the comparison, with exact-slice refinement and exact controlled evolution ($\sigma = 3$, 40 time sets), for $N = 4$ and $N = 6$ particles on $k = 4$ and $k = 6$ levels (eight and twelve qubits), always starting from the filled determinant of the smallest coarse space ($k_{\rm low} = N/2$). The pattern is the same throughout: the prolonged overlap falls to 0.78–0.80 at $k = 6$, refinement for $T = 30$ restores it to 0.997–0.999, and the UCCSD input, whose infidelity grows from 1.8×10^{-4} to 1.0×10^{-3} over the table, still enters the filter with a higher overlap than the refined state and leaves it with a lower infidelity at every cycle number. (At $N = 6$, $k = 4$ the ansatz is exact: three pairs on four levels are the particle–hole conjugate of one pair, for which UCCSD is complete.) The price of the UCCSD input is its circuit — 6 976 CNOTs and 92 parameters at $k = 6$, $N = 4$; 9 024 CNOTs and 117 parameters at $k = 6$, $N = 6$ — and the training of those parameters, which the refinement avoids at the cost of $2n_s(N^{\rm step}_{\rm CNOT} + N^{A}_{\rm CNOT})$ CNOTs with $N^{\rm step}_{\rm CNOT} = 732$ at $k = 6$. This is a controlled small-system demonstration, not a scaling study; a regime in which refinement wins would require a Hamiltonian for which the variational input is poor, and we do not find one within the pairing model at these sizes.

TABLE IV. The pipeline and the UCCSD comparator at other sizes ($g = 1$): coarse space the filled determinant at $k_{\rm low} = N/2$, exact-slice refinement for $T = 30$, exact controlled evolution with $\sigma = 3$, 40 time sets. ov_0 and p are the overlaps of the prolonged and refined states with the exact ground state, the UCCSD columns give the infidelity and energy error of the sector-level UCCSD state together with its parameter and CNOT counts, and the last four columns the acceptance-weighted infidelity after M rodeo cycles from the refined and from the UCCSD input. The register has $2k$ qubits (8 and 12) and the uncontrolled first-order Trotter step of Eq. (10) costs 296 CNOTs at $k = 4$ and 732 at $k = 6$.

(k, N)	dim	gap	ov_0	$p\,(T{=}30)$	$1-\mathcal{F}_{\rm UCCSD}$	$\Delta E_{\rm UCCSD}$	params	CNOTs	refined+rodeo		UCCSD+rodeo	
									$M{=}2$	$M{=}6$	$M{=}2$	$M{=}6$
$(4,4)$	70	1.823	0.868	0.9988	1.8×10^{-4}	1.4×10^{-3}	26	1 312	3.4×10^{-4}	2.3×10^{-5}	4.1×10^{-5}	1.4×10^{-6}
$(6,4)$	495	2.017	0.798	0.9976	7.6×10^{-4}	6.1×10^{-3}	92	6 976	6.3×10^{-4}	4.1×10^{-5}	1.7×10^{-4}	7.6×10^{-6}
$(4,6)$	28	1.779	0.894	0.9991	$<10^{-15}$	$<10^{-15}$	15	688	2.5×10^{-4}	1.7×10^{-5}	0	0
$(6,6)$	924	2.052	0.780	0.9972	1.0×10^{-3}	8.3×10^{-3}	117	9 024	7.2×10^{-4}	4.8×10^{-5}	2.4×10^{-4}	1.0×10^{-5}

TABLE V. Level occupation numbers n_p and energy of the prepared states at $k = 4$, $N = 4$, $g = 1$ ($E_{\rm ref} = 1$); the filtered states use the Trotterized filter of Table III ($\delta t = 0.25$) and are acceptance-weighted averages over 40 time sets; $a\,10^{-b}$ stands for $a\times10^{-b}$.

state	n_0	n_1	n_2	n_3	$E-E_{\rm ref}$	$1-\mathcal{F}$
exact	1.929	1.803	0.197	0.071	−0.3645	0
reference det.	2	2	0	0	0	$1.3\,10^{-1}$
UCCSD	1.930	1.806	0.194	0.070	−0.3630	$1.8\,10^{-4}$
refined ($T{=}30$)	1.908	1.795	0.205	0.092	−0.3549	$2.3\,10^{-3}$
refined+rodeo $M{=}2$	1.920	1.798	0.202	0.080	−0.3621	$5.3\,10^{-4}$
refined+rodeo $M{=}6$	1.923	1.801	0.199	0.077	−0.3637	$1.5\,10^{-4}$
UCCSD+rodeo $M{=}2$	1.924	1.801	0.199	0.076	−0.3637	$1.3\,10^{-4}$

a. Observables. A state-preparation method for nuclear structure is judged by the observables of the state it delivers, not only by its overlap. Table V lists the level occupation numbers $n_p = \sum_\sigma \langle n_{p\sigma}\rangle$ and the correlation energy $E-E_{\rm ref}$ of the states of Table III at $k = 4$, $N = 4$, $g = 1$ (for the filtered states, the acceptance-weighted average over the 40 time sets). The exact ground state depletes the two occupied levels by 0.07 and 0.20 particles. The UCCSD state reproduces the occupations to 3×10^{-3} and the correlation energy to 1.4×10^{-3}; the refined state at $T = 30$, whose infidelity is 2.3×10^{-3}, deviates by up to 0.02 in the occupations and by 10^{-2} in the correlation energy, and two rodeo cycles reduce both deviations by a factor 2–3, six cycles by a factor 4–5. Occupation numbers converge more slowly than the energy, as expected for a quantity that is linear rather than quadratic in the error of the state.

VI. BEYOND SENIORITY: ADDING A PARTICLE-HOLE INTERACTION

Everything above rests on a Hamiltonian that is, for the purpose of testing methods, almost too well behaved. The pairing force of Eq. (1) moves complete pairs and does nothing else; the seniority is conserved, the $N = 4$ ground state lives in a $\binom{k}{2}$-dimensional seniority-zero space, and an entire class of excitations is absent — there is no matrix element between the reference determinant and any one-particle–one-hole state. Realistic nuclear and electronic Hamiltonians do have that channel. The pure pairing model is by no means outside the reach of mean-field-based methods — the Richardson model has long served as a testing ground for particle-particle RPA and its self-consistent and renormalized variants [35, 36] and for the quasiparticle RPA [34] — but the particle-hole channel on which Tamm–Dancoff and particle-hole RPA are built is simply closed in it. To test the pipeline on a Hamiltonian with the central ingredients of a realistic two-body interaction we therefore add, following Chapter 4 of Ref. [22], one extra term to Eq. (1), keeping the level scheme, the particle number $N = 4$ and the resolutions $k = 2, 3, 4$ unchanged:

$$V_{\rm ph} = -\frac{f}{2}\sum_{p,q,r=0}^{k-1}\Big(a^\dagger_{p\uparrow}a^\dagger_{p\downarrow}a_{q\downarrow}a_{r\uparrow} + \text{h.c.}\Big). \qquad (12)$$

The triple sum contains three distinct processes. The $q = r$ pieces are the pairing term again and merely renormalize $g \to g + 2f$; the $q = p \neq r$ pieces carry a *single* particle from one level to another, a $1p$–$1h$ excitation of seniority two; and the pieces with p, q, r all different remove two unpaired particles and deposit a pair elsewhere, a seniority-two $2p$–$2h$ excitation. $V_{\rm ph}$ still conserves $N_\uparrow$ and $N_\downarrow$ separately, so S_z remains a good quantum number, but the seniority does not: at $k = 4$ the $S_z = 0$ sector has 36 determinants instead of the six pair configurations, and the compact one-qubit-per-pair encoding of the pairing model [23, 33] is no longer available. We keep the $2k$-qubit Jordan–Wigner register used throughout this work; it is the convenient general choice, not the only one — the conserved particle number and S_z still permit compressed encodings of the 36-dimensional block — but it is the register on which the gate counts below are meaningful. Acting once on the reference determinant at $k = 4$, the Hamiltonian now reaches eight $1p$–$1h$ and eight seniority-two $2p$–$2h$ determinants with matrix elements $\pm f/2$, in addition to the four seniority-zero $2p$–$2h$ states whose coupling grows from $g/2$ to $(g + 2f)/2$. Two consequences matter for

the pipeline. First, the filled determinant is no longer a stationary Hartree–Fock solution: its one-particle–one-hole matrix elements $\langle\Phi_i^a|H|\Phi_0\rangle = \pm f/2$ do not vanish (Brillouin's theorem holds for the self-consistent determinant, which we have not re-optimized), so the UCCSD *singles*, identically zero for the pairing force, switch on and absorb part of what an optimized reference would describe as mean-field relaxation rather than correlation. Second, the reference energy becomes $E_{\rm ref} = 2 - g - 2f$, and the quantity we call correlation energy below is measured from this unrelaxed reference. We use the strengths $f = 0.05$, 0.2 and 0.5 of Ref. [22] at $g = 1$, so that every exact number below can be cross-checked against its tables, and we apply the pipeline exactly as before — the only change is the Hamiltonian passed to each stage.

A. Exact and variational references

Table VI collects the reference energies, the exact ground-state energies and the UCCSD energies at $k = 3$ and $k = 4$. The $k = 4$ exact values reproduce Ref. [22] to all eight digits, now obtained in the full 70-dimensional $N = 4$ sector of the eight-qubit register rather than in the 36-dimensional $S_z = 0$ block (the ground state lies in that block for every f). As f grows the correlation energy grows much faster than the reference energy falls, from -0.36 at $f = 0$ to -1.69 at $f = 0.5$ (Fig. 14, left), the signature of a second, competing mean field. At $k = 3$ the UCCSD ansatz remains exact for every f, now with non-vanishing singles amplitudes (up to 0.09 at $f = 0.5$). At $k = 4$ the ansatz energy error grows by a factor of twenty across the scan, from 1.4×10^{-3} at $f = 0$ to 3.2×10^{-2} at $f = 0.5$, and the infidelity of the optimized state by the same factor, from 1.8×10^{-4} to 3.7×10^{-3}, tracking the weight of the exact ground state outside the seniority-zero space, which reaches 3.9% at $f = 0.5$ (Fig. 14, right). This is the first qualitative difference from the pairing model: the accuracy of the fixed single-reference ansatz, which did not depend on g before, now degrades with the strength of the pair-breaking channel. We note that these are the errors of the optimized Trotterized UCCSD circuit found by L-BFGS from a zero start; they document the accuracy of this preparation, not a proven expressivity limit of the ansatz.

TABLE VI. The pairing model with the particle-hole term of Eq. (12), $N = 4$, $g = 1$. $E_{\rm ref} = 1 - 2f$ is the energy of the reference determinant, FCI the exact N-sector ground state, UCCSD the statevector unitary coupled-cluster energy (identical to the gate-level circuit at $k = 3$), $\Delta E_{\rm UCCSD} = E_{\rm UCCSD} - E_{\rm FCI}$, $1 - \mathcal{F}_{\rm UCCSD}$ the infidelity of the optimized UCCSD state with respect to the exact ground state, and w_0 the weight of the exact ground state in the seniority-zero subspace. The $k = 4$ FCI column agrees with the corresponding table of Ref. [22].

k	f	FCI	UCCSD	$\Delta E_{\rm UCCSD}$	$1 - \mathcal{F}_{\rm UCCSD}$	w_0
3	0	0.794697	0.794697	$< 10^{-14}$	$< 10^{-14}$	1
3	0.05	0.650378	0.650378	$< 10^{-14}$	$< 10^{-14}$	0.9994
3	0.2	0.185803	0.185803	$< 10^{-14}$	$< 10^{-14}$	0.9938
3	0.5	−0.839538	−0.839538	$< 10^{-14}$	$< 10^{-14}$	0.9808
4	0	0.635548	0.636987	1.4×10^{-3}	1.8×10^{-4}	1
4	0.05	0.450582	0.453024	2.4×10^{-3}	3.0×10^{-4}	0.9988
4	0.2	−0.183485	−0.175303	8.2×10^{-3}	1.0×10^{-3}	0.9869
4	0.5	−1.691737	−1.659594	3.2×10^{-2}	3.7×10^{-3}	0.9605

B. Refinement with pairs broken

We prolong the $k = 2$ state — at $N = 4$ still a single determinant, now with energy $E_{\rm low} = E_{\rm ref}$ — into the $k = 4$ space and evolve along the path of Eq. (3) with the same shift rule $\mu = E_{\rm low} + 0.6$, for each f (Fig. 15 and Table VII). Two things change with f. The target state acquires seniority-two components that the prolonged determinant does not contain, and the effective pairing strength $g + 2f$ grows; both lower the initial overlap, from 0.868 at $f = 0$ to 0.600 at $f = 0.5$. The minimum gap along the path, however, is still set by the shift, $\Delta E_{\rm min} = 0.600$ for every f (the physical gap of $H_{\rm high}$ in fact *opens*, from 1.82 to 2.86), so the adiabatic time grows only because the state has further to travel: T^* rises from 16 to 32, and for every f the overlap exceeds 0.99 with the energy converging to FCI from above. The comparison with the coupling scan of Sec. IV D is telling: the pure pairing model at $g = 2$ — the value of $g + 2f$ at $f = 0.5$ — had $\mathrm{ov}_0 = 0.617$ and $T^* = 32$, almost exactly the numbers found here at $f = 0.5$. The refinement cost of the seniority-breaking model is thus governed by the renormalized pairing strength, and the genuinely new $1p$–$1h$ and pair-breaking channels add only a small correction to it. The sufficient conditions of Table II see the same picture from the other side: the time bound is insensitive to f because the minimum gap is set by the shift, while the Trotter-step bound grows with the number of Trotter groups that $V_{\rm ph}$ adds, from $\Lambda_2 = 13$ to 184. The $k = 3 \to 4$ chain of Sec. IV C, started from the gate-level UCCSD state whose singles now carry non-zero angles, buys the degradation back: it starts at overlap 0.851 even at $f = 0.5$ and reaches 0.99 at $T^* = 26$ instead of 32.

C. Rodeo filtering with a denser spectrum

The rodeo algorithm is the stage most exposed to the extra term, because its energy scan sees the whole spectrum of the eight-qubit Hamiltonian in the $N = 4$ sector, which the particle-hole term makes both denser and lower: 19 eigenvalues lie below $E = 4$ at $f = 0.5$ against 14 at $f = 0$. Figure 16 repeats the energy scan of Fig. 8 at the strongest coupling, $f = 0.5$, feeding the filter the resolution-refined ($T = 25$) state with $p = 0.980$ and, for contrast, the prolonged determinant with $p = 0.600$.

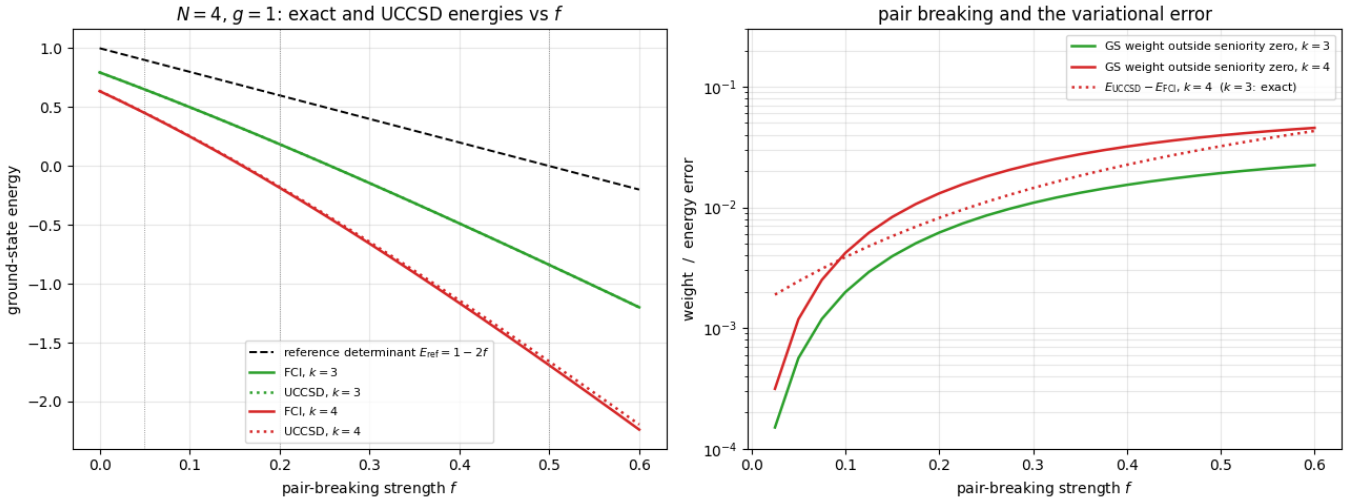


FIG. 14. The pairing model with the particle-hole term, $N = 4$, $g = 1$, as a function of the pair-breaking strength f. Left: energy of the reference determinant (dashed), exact ground-state energy (solid) and UCCSD energy (dotted) at $k = 3$ and $k = 4$; the vertical lines mark the strengths $f = 0.05, 0.2, 0.5$ of Table VI. Right: weight of the exact ground state outside the seniority-zero subspace (solid) and the UCCSD energy error at $k = 4$ (dotted; at $k = 3$ the ansatz is exact).

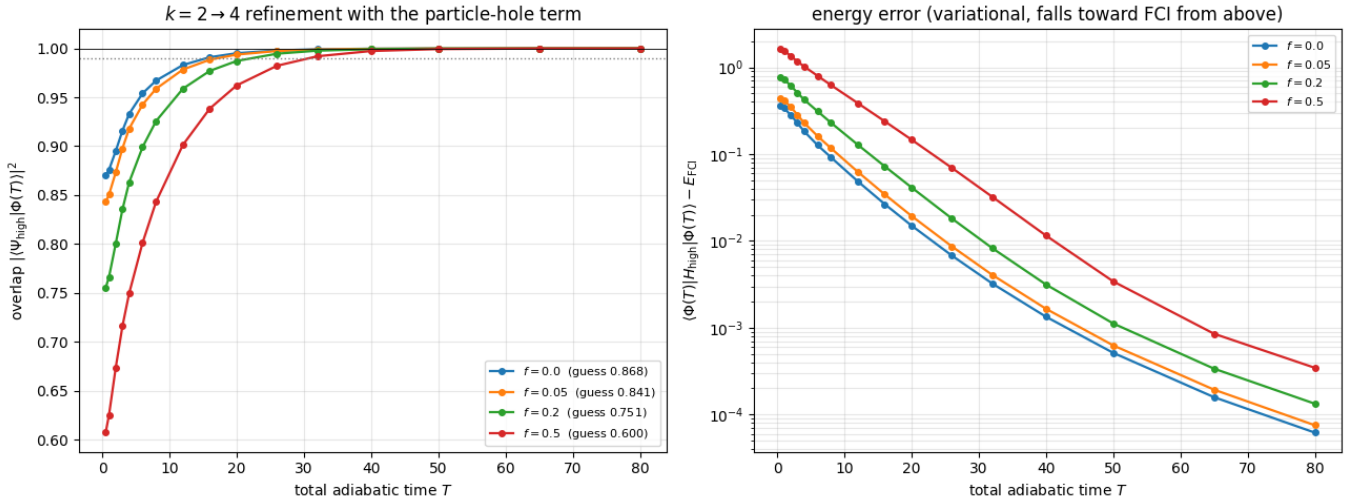


FIG. 15. Resolution refinement $k = 2 \to 4$ at $N = 4$, $g = 1$, for the pair-breaking strengths $f = 0, 0.05, 0.2, 0.5$. Left: overlap with the exact ground state versus the adiabatic time T (legend: initial overlap of the prolonged determinant; dotted line: the 0.99 target). Right: variational energy error $\langle\Phi(T)|H_{\rm high}|\Phi(T)\rangle - E_{\rm FCI}$ on a logarithmic scale.

The coarse scan resolves the ground state from the excited band — helped by the larger gap — and the fine scan locates the peak at $E = -1.6924$ against the exact $E_0 = -1.691737$; with the refined input the excited-state structure that the prolonged input shows above the 2^{-M} background is essentially gone. Fixing $E = E_0$ and tracking the fidelity with the number of cycles for all four strengths (Fig. 17), the infidelity falls geometrically along the $\frac{1-p}{p}2^{-M}$ reference and the acceptance settles at the input overlap: after ten cycles $1-\mathcal{F} = 2.0\times10^{-6}$, 2.9×10^{-6}, 4.6×10^{-6} and 4.5×10^{-5} for $f = 0$, 0.05, 0.2 and 0.5, with acceptances 0.998, 0.997, 0.994 and 0.980. The pair-breaking channels cost the pipeline one order of magnitude in final infidelity at the strongest coupling — recoverable with three more cycles — and nothing in acceptance beyond the lower input overlap.

D. The circuit-level budget with the particle-hole term

The pipeline itself is unchanged by $V_{\rm ph}$ — prolongation, the adiabatic path, the shift rule and the rodeo cycle were all written for the $2k$-qubit register — but the Hamiltonian that is Trotterized is not. Table IX lists the

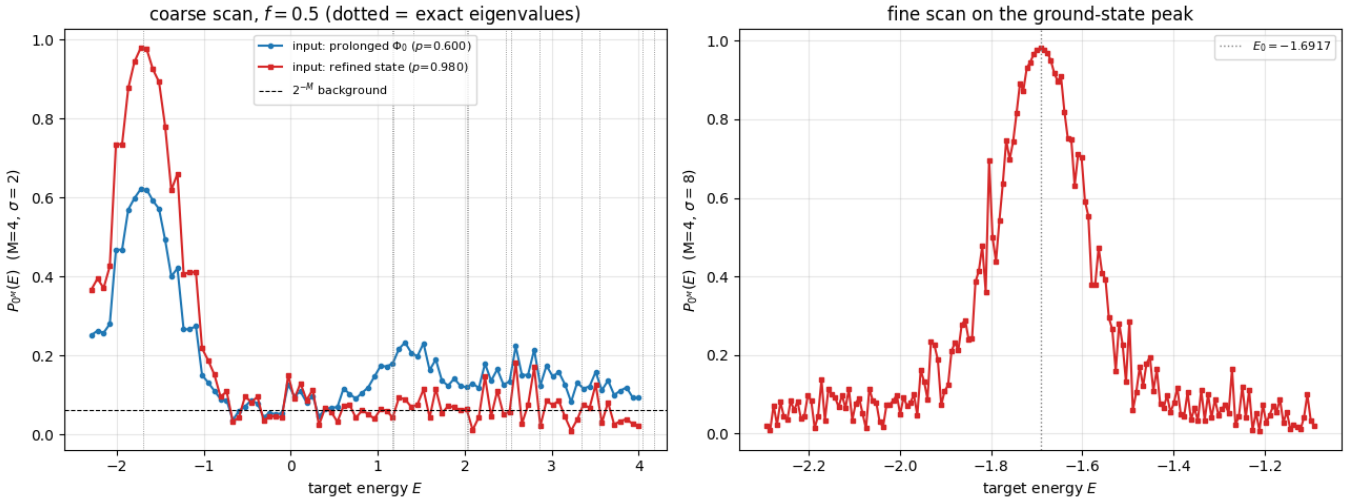


FIG. 16. Rodeo energy scan ($M = 4$) for the pairing model with the particle-hole term at $f = 0.5$. Left: coarse scan ($\sigma = 2$) from the prolonged determinant ($p = 0.600$) and from the resolution-refined state ($p = 0.980$); dotted vertical lines mark the exact eigenvalues, the dashed line the 2^{-M} background. Right: fine scan ($\sigma = 8$) on the ground-state peak, located at $E = -1.6924$ (exact $E_0 = -1.6917$).

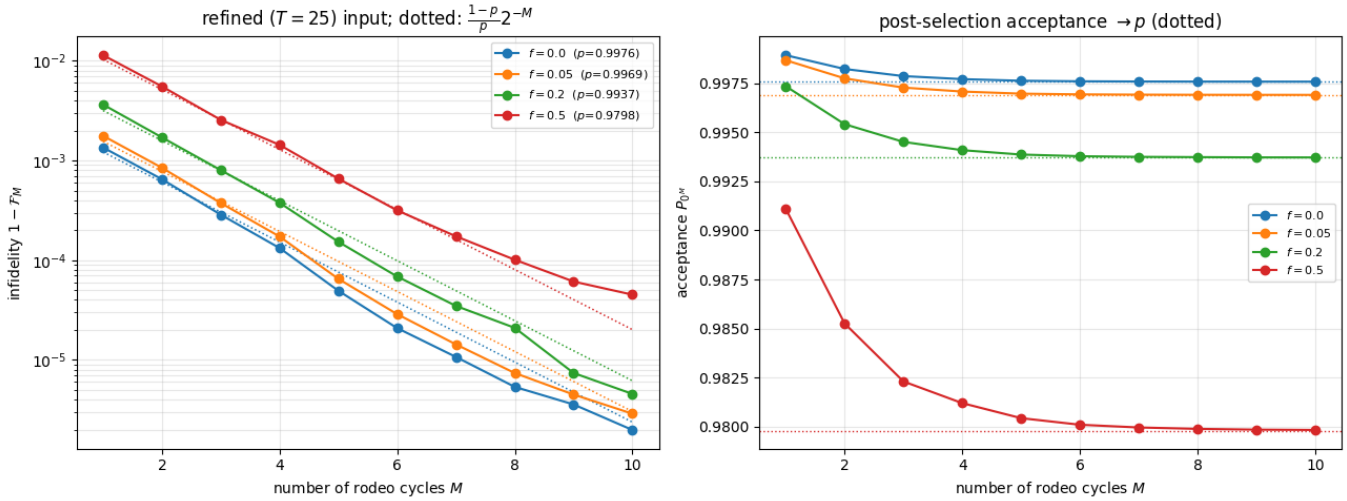


FIG. 17. Rodeo eigenstate preparation at $E = E_0$ with the particle-hole term, for $f = 0, 0.05, 0.2, 0.5$, from the resolution-refined ($T = 25$) input, averaged over 40 random time sets ($\sigma = 3$). Left: post-selected infidelity versus the number of cycles M (dotted: $\frac{1-p}{p}2^{-M}$). Right: the acceptance converges to the input overlap p (dotted).

Pauli-string content of H on $2k$ qubits. The particle-hole term multiplies the number of strings by a factor that grows with the resolution (1.8, 3.5 and 5.3 at $k = 2, 3, 4$; $61 \to 325$ at $k = 4$) and raises the largest Pauli weight from four to $2k$ (8 at $k = 4$), because a $1p$–$1h$ string drags a Jordan–Wigner parity chain across the register.

To extend the circuit-level analysis of Sec. V B we decompose $V_{\rm ph}$ into exactly exponentiable blocks in the same spirit as the pair-hopping terms. Its $q = r$ pieces are pairing terms and only rescale the rotation angles of the existing blocks ($g \to g + 2f$), at no cost. The remaining pieces, one per (p, q, r) with $q \neq r$, fall into two families: $2k(k-1)$ one-particle–one-hole blocks ($a^\dagger_{p\uparrow} n_{p\downarrow} a_{r\uparrow}$ + h.c. and the spin-flipped partner), each a sum of four Pauli strings, and $k(k-1)(k-2)$ pair-breaking blocks (eight strings each, like a double excitation). Within each block the strings mutually commute, so every factor $e^{-i\hat{T}\delta t}$ is realized exactly by its rotations and the product-formula error again comes only from the non-commutativity *between* blocks: 55 of them at $k = 4$ instead of seven. Two things follow, both quantified in Fig. 18 and Table VIII. First, the Trotter error at fixed step count grows only modestly — at $t = 1$, $k = 4$ the second-order error with sixteen steps is 2.1×10^{-3}, 3.8×10^{-3} and 7.5×10^{-3}

TABLE VII. Refinement to $k = 4$ ($N = 4$, $g = 1$, $\mu = E_{\rm low} + 0.6$) with the particle-hole term: gap of $H_{\rm high}$, minimum instantaneous gap along the path, initial overlap ov_0 of the prolonged state and adiabatic time T^* to reach overlap 0.99 (on the T grid of Fig. 6; the $T^* = 9$ of Sec. IV C for the $3 \to 4$ chain at $f = 0$ used a finer grid), for the gate-free $2 \to 4$ chain and for the $3 \to 4$ chain started from the gate-level UCCSD state.

			$2 \to 4$		$3 \to 4$	
f	$\mathrm{gap}(H_{\rm high})$	$\Delta E_{\rm min}$	ov_0	T^*	ov_0	T^*
0	1.823	0.600	0.868	16	0.964	8
0.05	1.943	0.600	0.841	20	0.955	12
0.2	2.289	0.600	0.751	26	0.921	16
0.5	2.860	0.600	0.600	32	0.851	26

for $f = 0$, 0.2, 0.5 — because the new blocks carry the small coefficient $f/2$. The Trotterized refinement reaches its exact-slice value with the same $n_s = 80$ second-order steps at $f = 0.2$ (overlap 0.9961 against 0.9969), and the rodeo cycle at $\delta t = 0.25$ keeps an acceptance of 0.96 with an infidelity of 3.8×10^{-4} after six cycles, essentially the pairing-model operating point. At $f = 0.5$ the step $\delta t = 0.25$ begins to fail — acceptance 0.84, the symptom identified in Sec. V B — and $\delta t = 0.125$ restores it to 0.97 with an infidelity of 6×10^{-4} against 5×10^{-4} for exact evolution. Second, the depth of one step grows ninefold: compiled block by block with the ladder rule of Eq. (10), a first-order step costs 2728 CNOTs instead of 296 (348 rotations instead of 60; 3424 against 416 once controlled), the $1p$–$1h$ family contributing 640 and the pair-breaking family 1792 CNOTs; the coarse term is unchanged (14 CNOTs, the coarse space is still one determinant). The budget per accepted preparation therefore scales almost exactly with the step cost: at the pairing-model operating point ($n_s = 80$, $M = 2$, $\delta t = 0.25$) it is 6.5×10^5 CNOTs at $f = 0.2$ against 7.2×10^4 for the pure pairing force, for an infidelity of 1.0×10^{-3}, and at $f = 0.5$ the finer rodeo step brings the total to 8.3×10^5 CNOTs for 5.3×10^{-3}. The price of a more realistic interaction thus appears where it should, in the depth of each Trotter step of the Hamiltonian evolution, and not in the structure, the operating point or the convergence of the pipeline. Measured against the right comparator, however, the extra depth buys no accuracy advantage: the actual infidelity of the UCCSD state is 1.0×10^{-3} at $f = 0.2$ and 3.7×10^{-3} at $f = 0.5$, so the hybrid at $M = 2$ is tied with the *unfiltered* variational state at $f = 0.2$ and behind it at $f = 0.5$, and the same filter applied to the UCCSD input (last column of Table VIII) reaches 5.1×10^{-4} and 2.0×10^{-3} at $\delta t = 0.25$, 2.6×10^{-4} and 9.6×10^{-4} at $\delta t = 0.125$, for 2.0–3.8×10^5 CNOTs per accepted preparation — a factor two to three below the hybrid in both accuracy and cost. The seniority-breaking term thus confirms the structural robustness of the pipeline and, at the same time, the conclusion of Sec. V C.

VII. DISCUSSION AND OUTLOOK

The pairing model lets us exhibit the complete refinement-plus-filter chain with every stage benchmarked against exact results: a gate-level UCCSD coarse solver, a gate-free prolongation with a counted conditioned coarse term, a gapped adiabatic refinement whose extracted time grows linearly with the inverse of the shift-controlled path gap over the range studied, and a rodeo filter whose ensemble infidelity follows Eq. (9) — all validated a second time at the circuit level with product formulas. The two algorithms are complementary in the intended sense: refinement raises the overlap from 0.87 to 0.998, and the filter removes the adiabatic tail geometrically. The coupling scan adds a practical lesson: at strong pairing the limiter is not the path gap but the quality of the prolonged starting state. The particle-hole extension of Sec. VI shows that none of this depends on the seniority symmetry that makes the pairing model convenient: with the one-particle–one-hole channel open, the same stages run unchanged, the path gap is still set by the shift, the refinement cost is governed by the renormalized pairing strength $g + 2f$, and the filter converges geometrically for every strength studied; the cost of the more realistic interaction shows up in the Pauli-string content of the Hamiltonian (61 → 325 strings at $k = 4$) and hence in the depth of each Trotter step (296 → 2728 CNOTs).

The central quantitative finding, however, is negative for the refinement stage as a gate-saving device. Compared on the same observable — the ground-state infidelity — and with the same Trotterized filter, refinement followed by the rodeo algorithm is tied with filtering the unrefined prolonged determinant and is beaten, in accuracy and in cost per accepted preparation, by filtering the full-space UCCSD state, for the pure pairing force as well as with the particle-hole term and at every size we can validate exactly (Tables III, IV and VIII). The reason is simple: in these systems a single-reference variational circuit with a few tens of parameters already has an overlap above 0.998, and an overlap of 0.87 costs the filter only a few extra cycles, whereas an adiabatic path of length $T^* \sim$ 10–30 at a Trotter step of several hundred CNOTs costs tens of thousands. What the refinement offers in exchange is that it is optimizer-free and measurement-free during preparation, and that its cost is set by the Hamiltonian evolution rather than by the trainability of a landscape. Whether that trade pays off is a question about regimes we cannot reach with exact validation — larger spaces, interactions for which single-reference ansätze fail, or hardware on which the training measurements of a variational loop dominate — and we regard the present work as the controlled baseline against which such claims should be measured, not as evidence for them.

a. Relation to symmetry-restoration and subspace approaches. The results invite a comparison with the pairing-model work of Lacroix and collaborators summa-

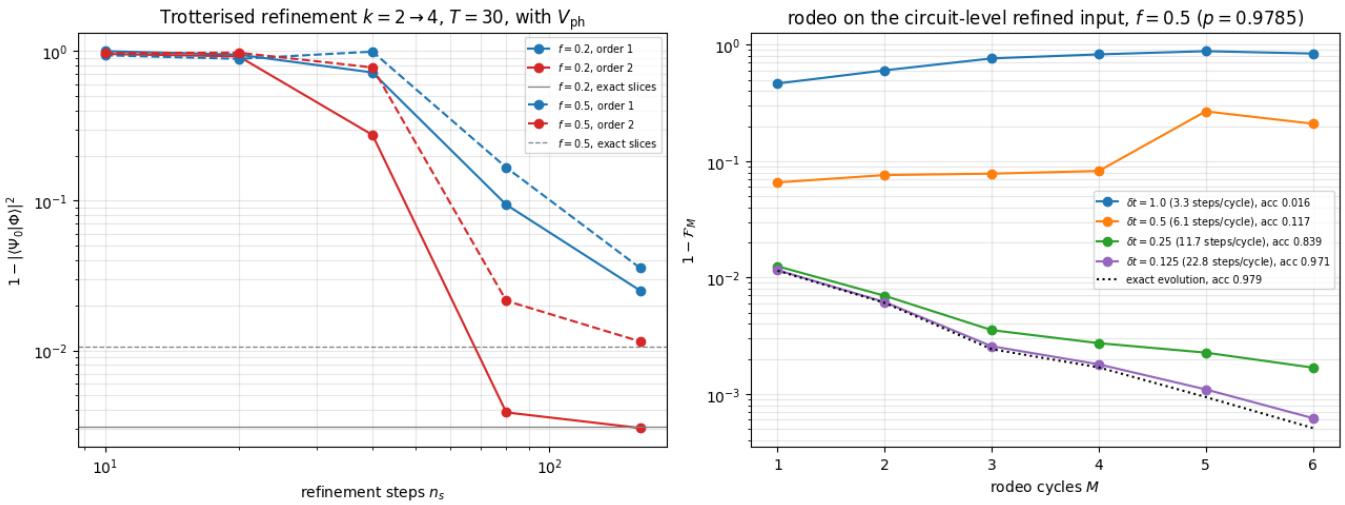


FIG. 18. The pipeline at the circuit level with the particle-hole term ($k = 4$, $N = 4$, $g = 1$). Left: infidelity of the Trotterized refinement $k = 2 \to 4$ ($T = 30$) versus the number of product-formula steps n_s for first- and second-order formulas at $f = 0.2$ (solid) and $f = 0.5$ (dashed); horizontal lines mark the exact-slice refinement. Right: rodeo infidelity versus cycles at $f = 0.5$ on the circuit-level refined input ($n_s = 80$, second order, $p = 0.9785$) with the controlled evolution Trotterized at fixed step sizes δt; the legend gives the mean number of steps per cycle and the six-cycle acceptance.

TABLE VIII. Gate budget at $k = 4$ ($N = 4$, $g = 1$) with the particle-hole term, per accepted preparation, at the operating point of Table III ($n_s = 80$ second-order refinement steps at $T = 30$, $M = 2$ rodeo cycles, $\sigma = 4$) for two rodeo step sizes δt. Step costs are the block-by-block CNOT counts (uncontrolled / controlled); the refinement column includes the 14-CNOT coarse term per half-step; the rodeo column uses the mean number of Trotter steps over the two cycles, doubled for the symmetric formula. The last two columns give the infidelity of the hybrid and, for comparison, the infidelity of the unfiltered UCCSD state (1 312 CNOTs, 26 parameters, the same circuit for every f) and of the UCCSD state passed through the same two-cycle filter, with its CNOTs per accepted preparation in parentheses. The first row is the pure pairing model of Table III.

f	δt	step (unctrl./ctrl.)	refinement	rodeo	per run	P_M	per accepted	$1 - \mathcal{F}$: hybrid / UCCSD	UCCSD+rodeo (CNOTs)
0	0.25	296 / 416	49 600	22 000	71 600	0.994	7.2×10^4	5.3×10^{-4} / 1.8×10^{-4}	1.3×10^{-4} (2.3×10^4)
0.2	0.25	2728 / 3424	438 720	190 603	629 323	0.972	6.5×10^5	1.0×10^{-3} / 1.0×10^{-3}	5.1×10^{-4} (2.0×10^5)
0.2	0.125	2728 / 3424	438 720	375 499	814 219	0.996	8.2×10^5	8.1×10^{-4} / 1.0×10^{-3}	2.6×10^{-4} (3.8×10^5)
0.5	0.25	2728 / 3424	438 720	190 603	629 323	0.891	7.1×10^5	6.1×10^{-3} / 3.7×10^{-3}	2.0×10^{-3} (2.1×10^5)
0.5	0.125	2728 / 3424	438 720	375 499	814 219	0.979	8.3×10^5	5.3×10^{-3} / 3.7×10^{-3}	9.6×10^{-4} (3.8×10^5)

TABLE IX. Pauli-string content of the Hamiltonian on $2k$ qubits: number of strings and largest Pauli weight, for the pairing model and for the pairing model with the particle-hole term ($f \neq 0$; the counts do not depend on the value of f). The $k = 2, 3$ rows agree with Ref. [22].

k	qubits	pairing	pairing + V_{ph}
2	4	15 (weight 4)	27 (weight 4)
3	6	34 (weight 4)	118 (weight 6)
4	8	61 (weight 4)	325 (weight 8)

rized in the Introduction. Both programmes start from a cheap, imperfect trial state and sharpen it; the difference is what is sharpened and how. In the symmetry-restoration route the trial state breaks a symmetry on purpose and the projector — phase estimation on the symmetry operator [23], an oracle [28] or a measurement-side projection [31] — removes the unwanted sectors, after which a Krylov or phase-estimation stage refines the energy [26]. Our coarse state conserves particle number and spin by construction; what it lacks is *resolution*, and the refinement supplies it along a gapped path while the rodeo cycle filters in *energy* rather than in a symmetry quantum number. The two filters are structurally the same resource — one ancilla, a controlled unitary and a post-selected measurement — and they compose: a symmetry-projected input would enter our pipeline exactly as the UCCSD state does. The quantum subspace methods of Refs. [30, 32, 33] are complementary in a different way: they trade coherent depth for measurements and a classical generalized eigenvalue problem, and their accuracy is bounded by the quality of the states that span the subspace, whereas the rodeo filter converges geometrically in depth but requires the controlled evolution. Two observations of that work bear on ours, with the lim-

its of an energy filter kept in mind. Ref. [47] documents that ADAPT-VQE with a symmetry-breaking pool can collapse to unphysical energies on degenerate pairing systems. A downstream energy filter can repair such an input only insofar as the input retains a nonzero overlap with the target: it cannot recover a state of zero overlap, cannot move weight between particle-number sectors that the evolution conserves, and cannot separate exactly degenerate states by energy alone, so a symmetry-broken ADAPT state would have to be symmetry-projected before, not instead of, the filter. Ref. [33] shows that the accuracy of the one-body Green's function of a small superfluid system, computed with the pure pairing Hamiltonian across the normal-to-superfluid transition driven by the pairing strength, is controlled by the quality of the even-N ground state; a filtered state of the kind prepared here is a possible input to that construction, and the particle-hole term of Sec. VI — which is not part of the model of Ref. [33] — is the ingredient that forces one out of the pair-to-qubit encoding used there.

b. Relation to nucleation. The nucleation method of Ref. [9] is the closest relative of the refinement stage, and the comparison is instructive in both directions. Both start from a small problem solved exactly on a classical computer and compiled into a circuit, and both reach the target through a sequence of Trotterized adiabatic interpolations with the $\sin^2(\pi s/2)$ schedule of Eq. (3), each of which changes the Hamiltonian only locally: nucleation switches on the couplings of newly added lattice sites, initialized in $|+\rangle$, to the existing bulk; refinement switches off the conditioned coarse term A of Eq. (4) and switches on the matrix elements of H_{high} that connect the occupied coarse levels to the appended ones, initialized empty in $|0\rangle$. Nucleation grows the domain at fixed resolution, refinement the resolution at fixed domain, and the two compose in principle: a lattice or shell-model problem could be nucleated to its physical size in a coarse basis and then refined to the full basis, or the reverse, with the same Trotterized machinery and the same filter at the end.

What Ref. [9] adds to the present assessment is twofold. First, it provides the rigorous sufficient conditions on the total time and on the Trotter step count of each stage quoted in Sec. III C, and shows on a twelve-site transverse-field Ising chain that they overestimate the resources actually needed by orders of magnitude: an overlap of 0.997 is reached with $T \simeq 40$ and some 2×10^3 Trotter steps in total, against bounds of $T \geq 2.6\times10^3$ and $N \geq 2\times10^{10}$. Section IV E makes the same comparison for the refinement path: the time bound of Eq. (5) overshoots the empirical T^* by a factor 1.4–3.2×10^2 and the step bound overshoots the steps actually needed by a factor 2×10^2–3×10^3 (Table II), because the bound charges the full norm of $B-A$ where the dynamics only pays for the matrix elements that couple the tracked state to its neighbors; the distance between bound and practice is the room in which the inverse-gap scaling of Ref. [7] operates. Second, and more directly relevant to our negative finding, Ref. [9] introduces *optimized nucleation*: the Trotterized growth circuit is kept as the ansatz structure and its Trotter angles as the initial parameters, and a small number L of layers per stage — far below the Trotter step count of the non-variational version — is then optimized layer by layer to minimize the energy. On a 5×5 transverse-field Ising model this reaches a lower energy error at fewer two-qubit gates than converting a matrix-product state to a circuit, even when the latter is itself re-optimized. The same construction is available for refinement: the n_s slices of Eq. (3) compiled in Sec. V B form a parameterized circuit whose angles are fixed by $(s,\delta t)$, and an *optimized refinement* with a few layers per stage would trade the one asset the adiabatic path has over the UCCSD input in Table III — freedom from classical optimization and training measurements — for a gate count that could fall well below the 4.96×10^4 CNOTs of the $T=30$ path. Whether a warm-started, physically structured ansatz of this kind escapes the trainability problems of a generic one is the question Ref. [9] raises for lattices; the pairing model, whose optimum is known exactly at every resolution, is the place to settle it, and we list it among the extensions below.

c. Hardware requirements. The Trotterized pipeline of Sec. V B quantifies the dominant coherent error of an ideal device, and the requirements it implies are explicit: $2k+1$ qubits (9 at $k=4$, 13 at $k=6$, 17 at $k=8$); a mid-circuit measurement with reset and reuse of one ancilla; arbitrary-angle R_z rotations, 160 of them per UCCSD preparation and 60 (pairing) to 348 (with V_{ph}) per Trotter step; and a coherent budget per accepted preparation that ranges from 2×10^4 two-qubit gates for the cheapest route of Table III (UCCSD input, two cycles) to 10^5 for the refined input with six cycles, and to 10^6 with the particle-hole term (Table VIII), on top of the energy scan and the repetitions needed to estimate acceptance probabilities. At the two-qubit error rates demonstrated in large-scale experiments today — of order 10^{-2} on the superconducting processor of Ref. [58] and $(0.8$–$2)\times10^{-3}$ on trapped-ion processors [59, 60] — a 10^4-gate run suffers of order ten faults even on the best of these and a 10^5-gate run of order one hundred; none of these budgets is executable with physical qubits without substantial error mitigation, and qubit counts alone say nothing about a device's suitability. Early logical-qubit demonstrations on neutral-atom arrays have executed circuits on tens of logical qubits with error detection and, in part, error correction and mid-circuit readout [61, 62], and the rodeo algorithm has been run in two-qubit form on trapped-ion and superconducting hardware [6, 50]; a logical-depth implementation of the budgets above remains to be demonstrated. A proportionate near-term target is therefore a mitigated $k=3$ demonstration: the $2\to3$ chain of Sec. IV B on seven qubits (six system qubits plus the ancilla), whose $N=4$ coarse state is a single determinant, or the $3\to4$ chain of Sec. IV C on nine qubits, with a short refinement schedule and one or two rodeo cycles at

coarse δt, i.e. 10^3–10^4 two-qubit gates, where the collapse of the acceptance under a corrupted evolution (Sec. V B) doubles as a hardware diagnostic. All-to-all connectivity absorbs the Jordan–Wigner ladders without SWAP overhead; on square or heavy-hex lattices the weight-4 pair-hopping rotations act on adjacent qubits, which limits but does not eliminate routing costs, and the weight-$2k$ strings of $V_{\rm ph}$ do not. The full $k = 4$ budget is a workload for the first generation of logical-qubit machines such as the neutral-atom system planned by the QuNorth initiative [57], and Tables III and VIII are offered as its specification.

d. Concluding remarks and perspectives. The present results establish, on a fully controlled example, how resolution refinement followed by the rodeo algorithm behaves at the circuit level — every stage compiled, counted and validated against exact results — and what it is worth: it prepares the correlated ground state to the accuracy of the Trotterized evolution without any variational optimization, at a gate cost that, for the pairing model at these sizes, exceeds that of filtering a cheap variational input. Three extensions would sharpen the assessment. First, comparators: an ADAPT-VQE or pair-adapted input, an optimized refinement in the sense of Ref. [9] — the Trotterized path of Eq. (3) as a warm-started variational circuit with a few layers per stage — and a jointly optimized choice of refinement time and cycle number along the cost–fidelity curves of Fig. 13, with statistical uncertainties from larger time-set ensembles. Second, Hamiltonians for which single-reference variational inputs fail — shell-model interactions with competing deformations, or the strong-coupling regime of Sec. IV D treated with a nontrivial coarse space — which is where the trainability argument for refinement would be tested rather than asserted. Third, hardware: a mitigated seven- or nine-qubit demonstration on present devices, where the acceptance statistics of the filter double as a diagnostic of the compiled evolution, followed by the full $k = 4$ budget on an early logical-qubit machine. The pairing model, exactly solvable at every resolution, remains the calibration standard for all three.

CODE AVAILABILITY

The complete implementation — pairing Hamiltonian with and without the particle-hole term, gate-level UCCSD–VQE, prolongation, adiabatic refinement, and the ancilla-based rodeo algorithm — is provided as a tested Python package and a single Jupyter notebook that reproduces every figure in this article. Give GitHub link when everything is ready.

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